

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Coherence (Part 5)

Instructor: Prof. Joshua San Miguel

Some slides adopted from Greg Steffan, Natalie Enright Jerger and Mikko Lipasti

Last Lecture

- Directory-based coherence – MSI
- Directory structure

This Lecture

- Directory-based coherence – MESI
- Coherence optimizations
- Virtual networks

Directory-Based Coherence – MESI

Modified-Exclusive-Shared-Invalid (MESI) Protocol

- Write-back, write-allocate caches

Processor to cache (i.e., *mandatory* requests):

- *Load*: Read block in cache
- *Store*: Modify block in cache
- *Evict*: Remove block from cache

Cache to directory:

- *GetS*: Request read access to block
- *GetM*: Request write (exclusive) access to block
- *PutS*: Clean eviction notification
- *PutM*: Dirty eviction write-back
- ***PutE*: Exclusive clean eviction notification**

Directory to cache:

- *Fwd-GetS*: Tell owner to send block to sharer
- *Fwd-GetM*: Tell owner to send block to writer
- *Put-Ack*: Notify cache of eviction receipt
- *Inv*: Invalidate sharer cache
- *Data*: Send block to requester
- ***DataE*: Send block to exclusive requester**

Cache to cache:

- *Inv-Ack*: Notify cache of invalidation receipt
- *Data*: Send block to requester

Directory-Based Coherence – MESI

Modified-Exclusive-Shared-Invalid (MESI) Protocol

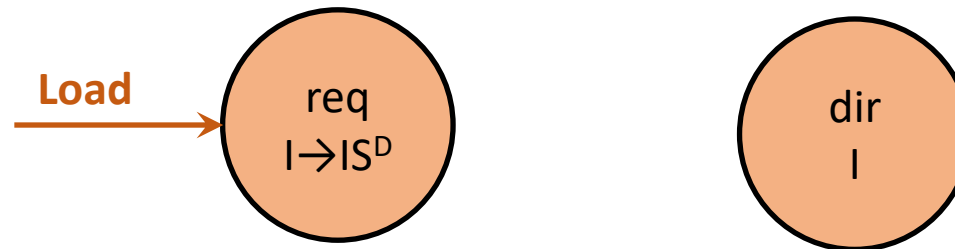
Load?



Directory-Based Coherence – MESI

Modified-Exclusive-Shared-Invalid (MESI) Protocol

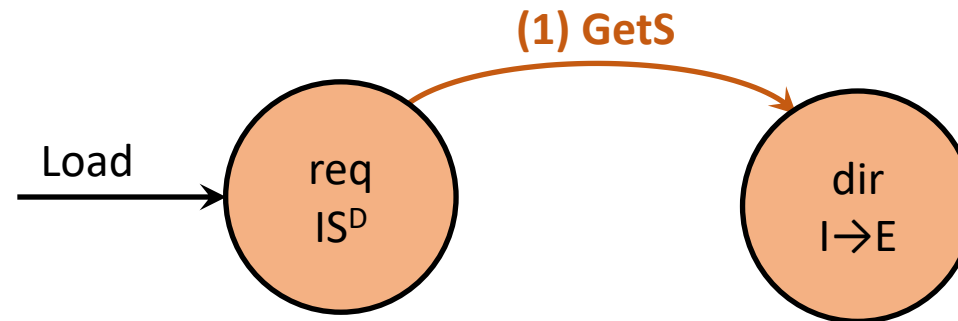
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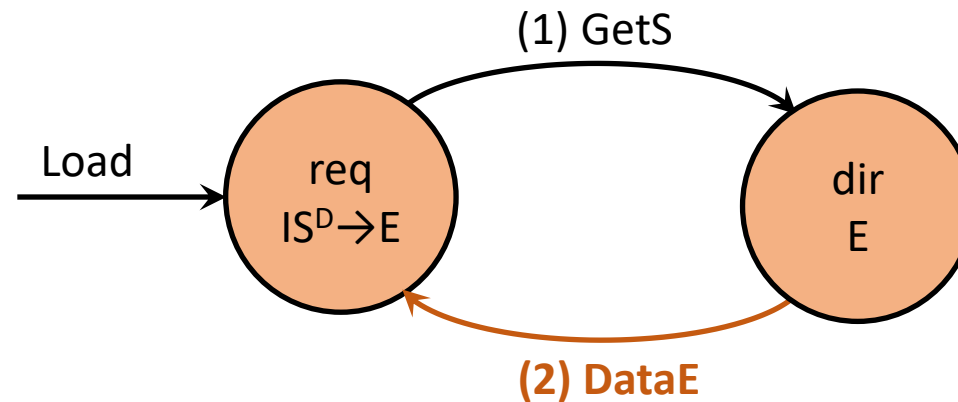
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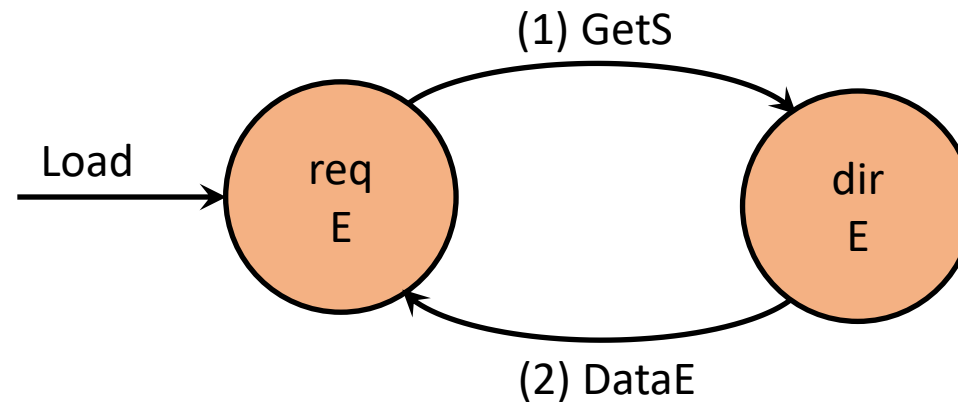
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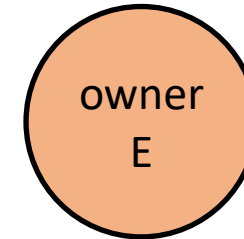
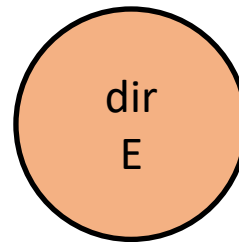
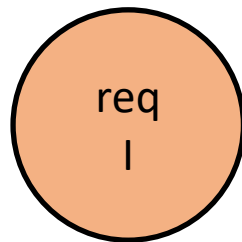
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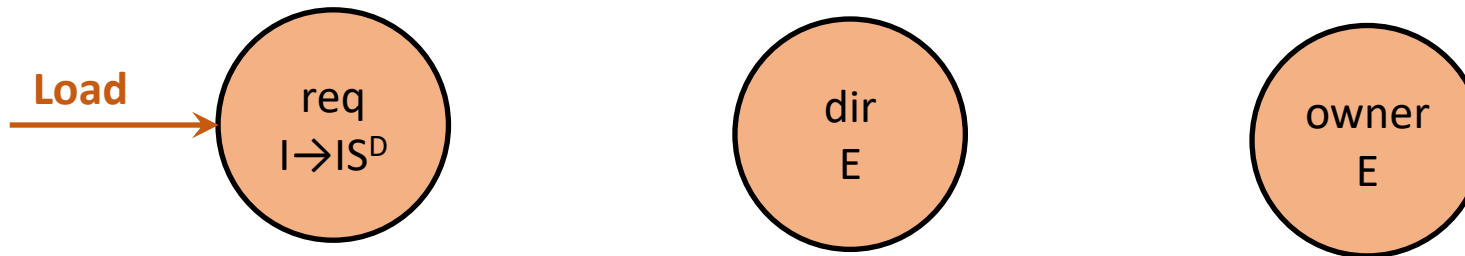
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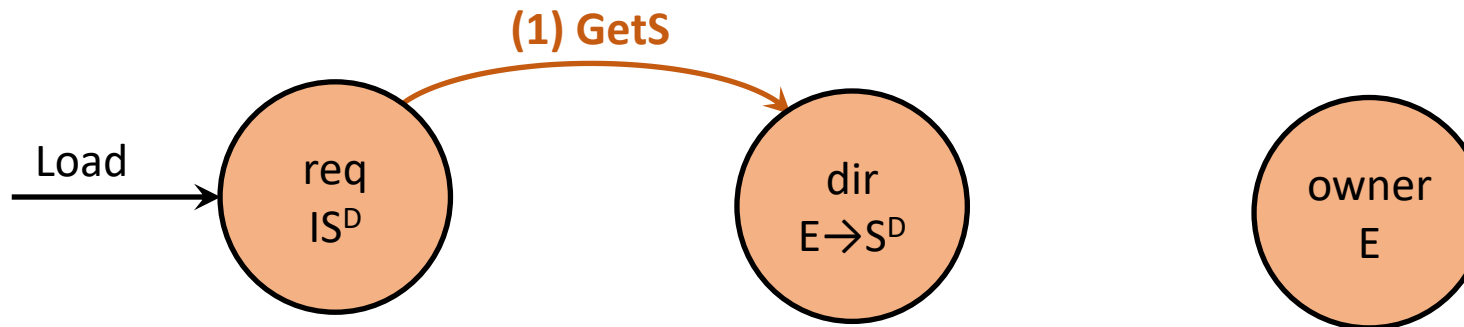
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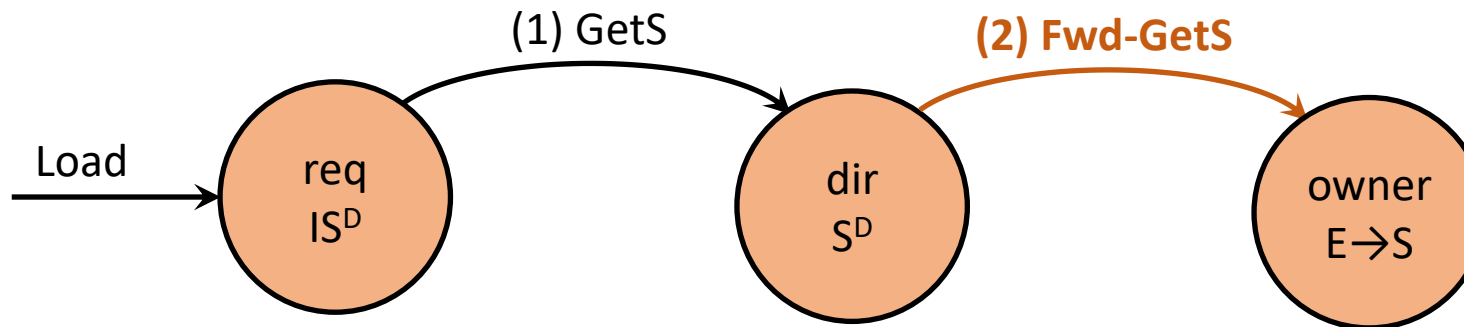
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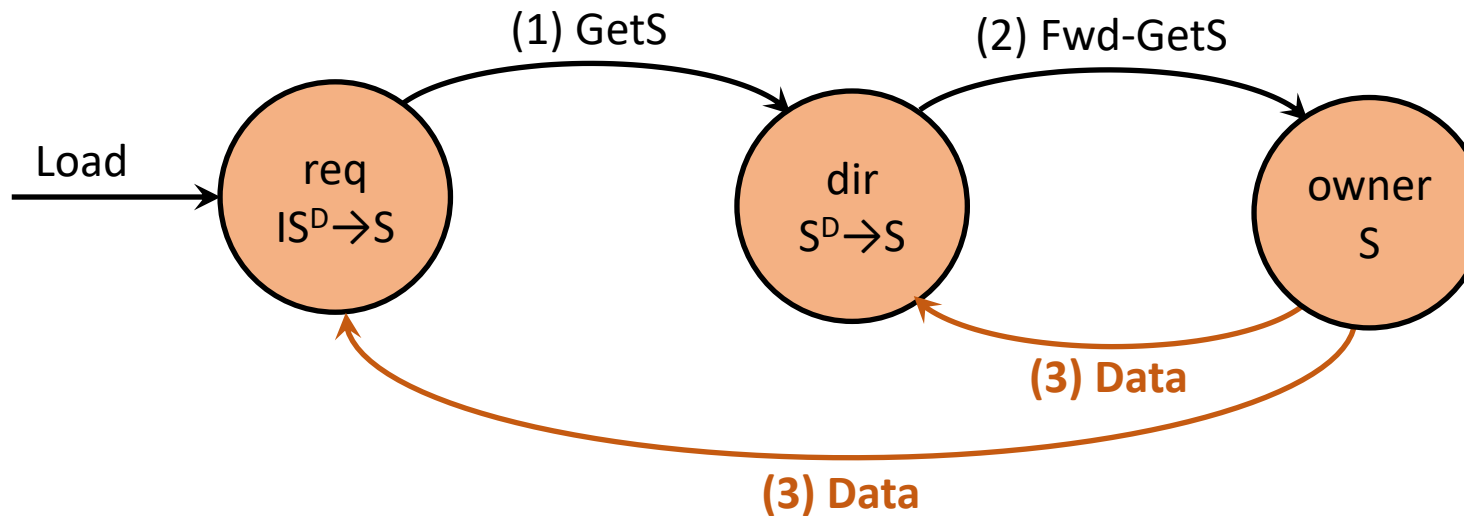
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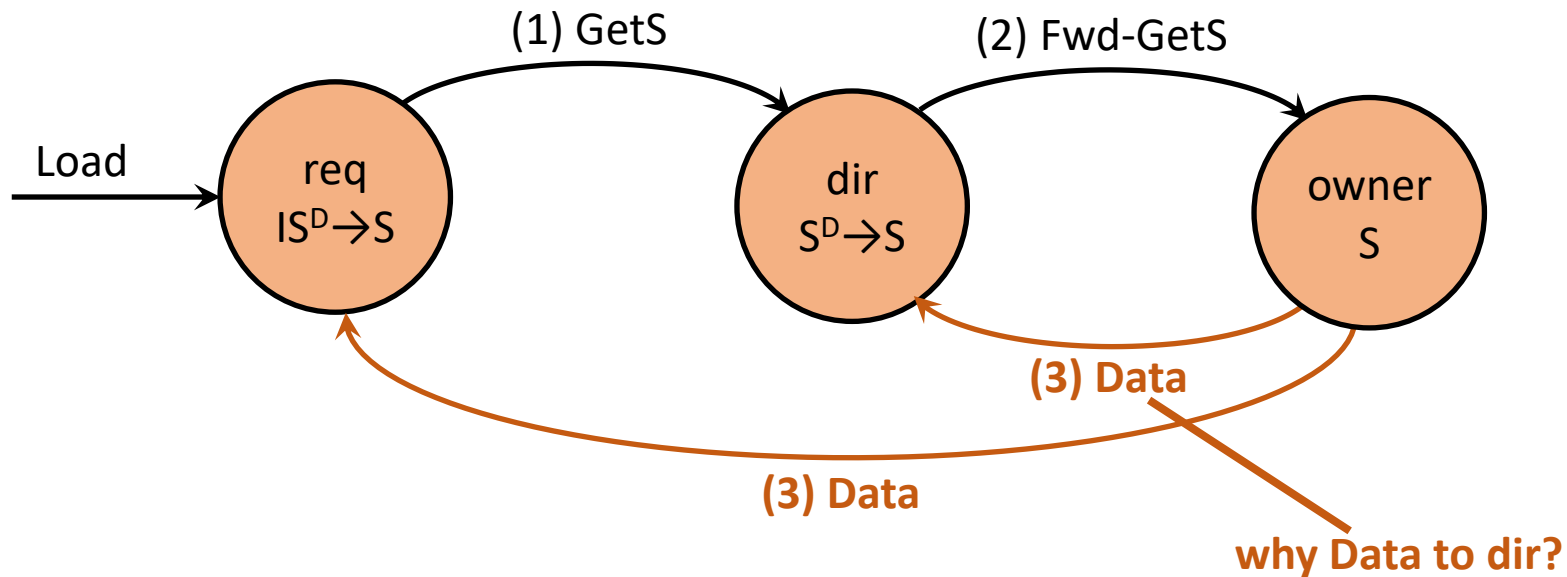
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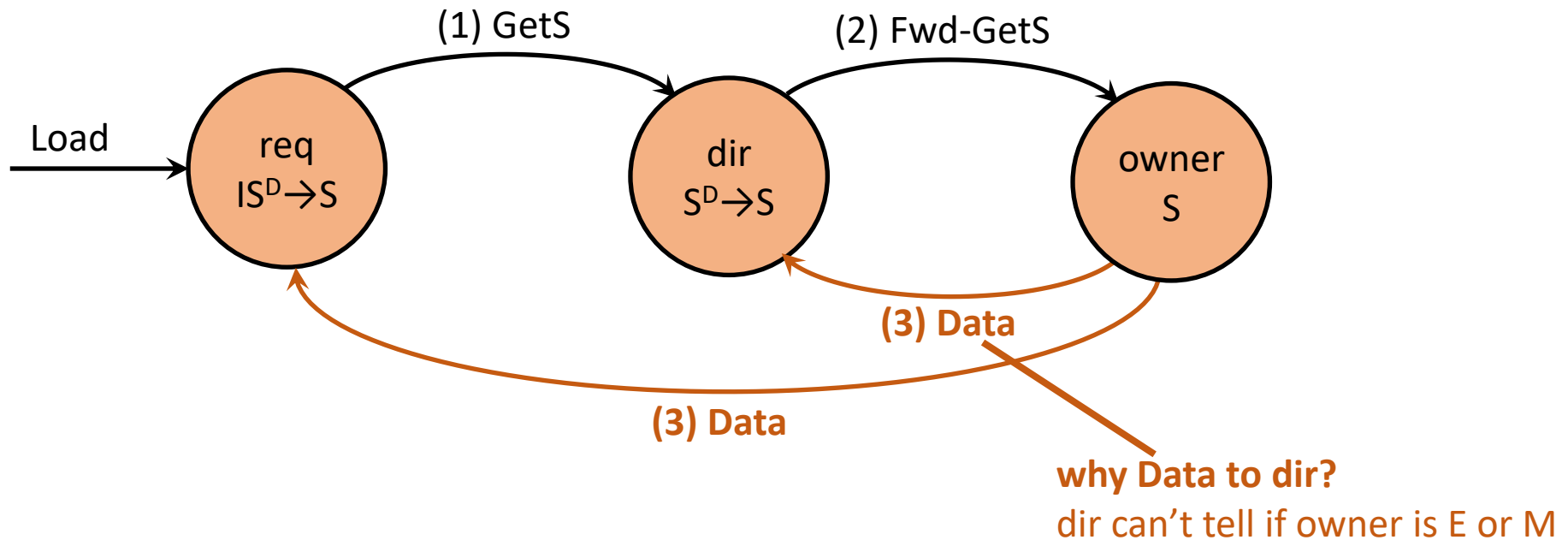
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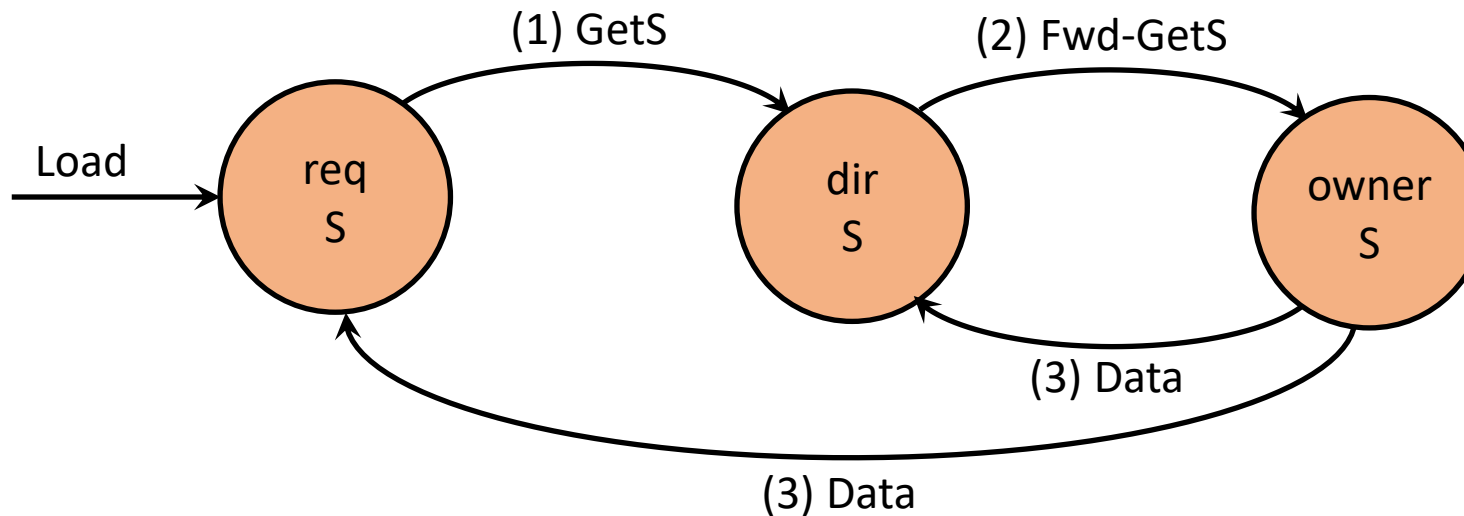
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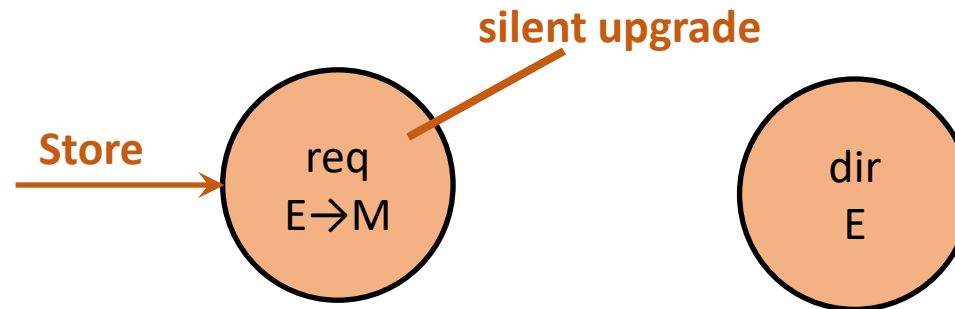
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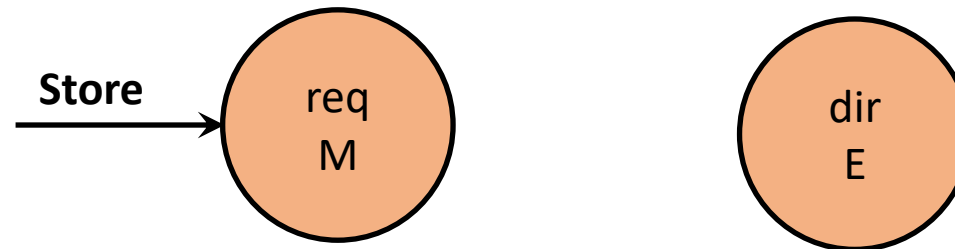
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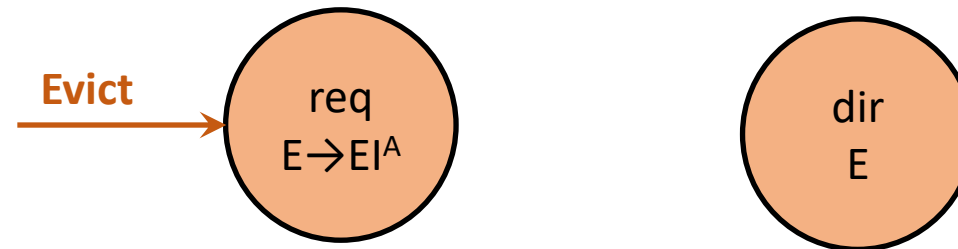
Evict?



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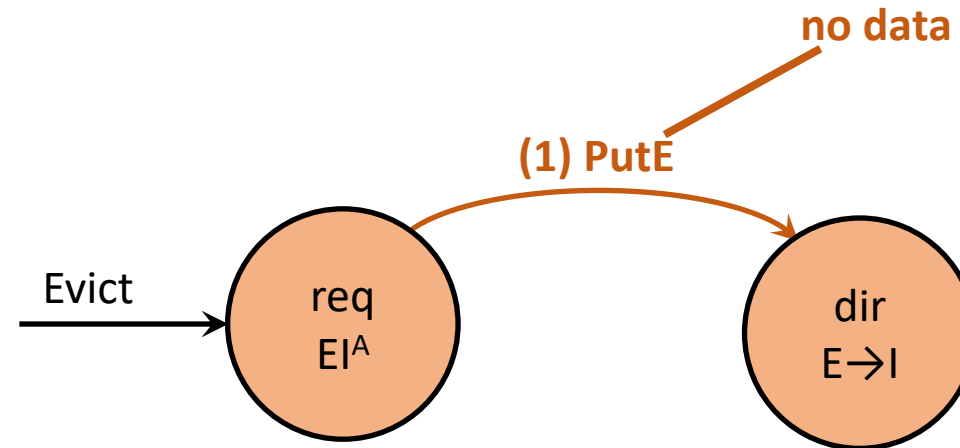
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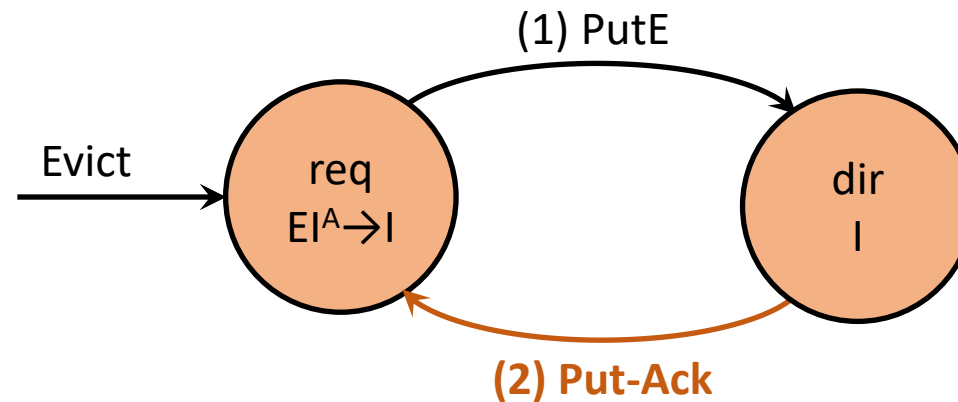
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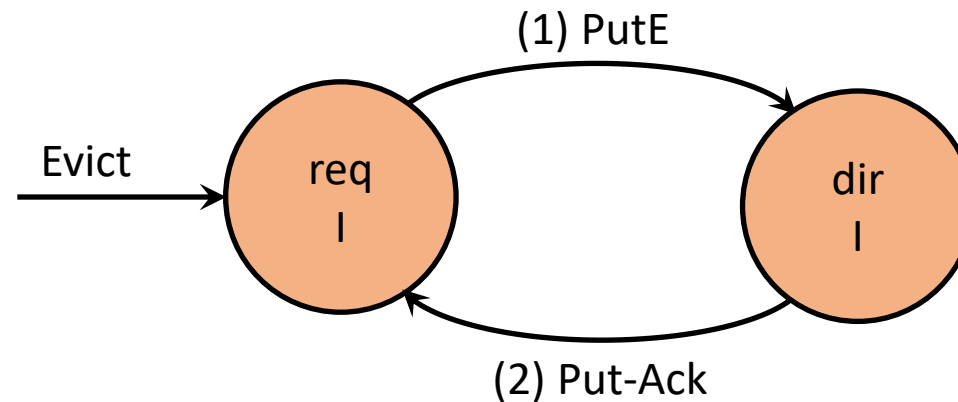
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Directory-Based Coherence – MESI

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Evict?



Directory-Based Coherence – MESI

Cache Controller

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	DataE from dir	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}											
IS ^D	stall	stall	stall			stall			→ S		→ S		
IM ^{AD}	stall	stall	stall	stall	stall				→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall							ack-- → IM ^A	→ M
...	...												
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I								
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SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I						
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Directory-Based Coherence – MESI

Directory Controller

Current State	Event								
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	PutE (owner)	PutE (not owner)	Data
I	Data to req → S	Data to req → M	Put-Ack to req → I	Put-Ack to req → I		Put-Ack to req → I			
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Put-Ack to req → S			
E									
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M			
S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



Directory-Based Coherence – MESI

Directory Controller

Current State	Event								
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	PutE (owner)	PutE (not owner)	Data
I	Data to req → S	Data to req → M	Put-Ack to req → I	Put-Ack to req → I		Put-Ack to req → I			
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Put-Ack to req → S			
E									
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M			
S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



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Directory-Based Coherence – MESI

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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



Directory-Based Coherence – MESI

Directory Controller

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Directory-Based Coherence – MESI

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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



Directory-Based Coherence – MESI

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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



Directory-Based Coherence – MESI

Directory Controller

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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D			Data to mem → S



Directory-Based Coherence – MESI

Directory Controller

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Directory-Based Coherence – MESI

Directory Controller

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	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	PutE (owner)	PutE (not owner)	Data
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Directory-Based Coherence – MESI

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Directory-Based Coherence – MESI

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Directory-Based Coherence – MESI

how is it possible to receive PutE (not owner)
while in E state?

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S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D		Put-Ack to req → S ^D	Data to mem → S

All of the above happens before original owner's PutE arrives at dir



Directory-Based Coherence – MESI

Directory Controller

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Directory-Based Coherence – MESI

why not transition to E upon receiving
second last PutS?

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M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Not useful if last sharer doesn't know; Dir would need to send message				
S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D					



Directory-Based Coherence – Optimizations

Sharer prediction:

- Prediction in coherence protocols [S. S. Mukherjee and M. D. Hill, ISCA 1998]
- Instruction-based prediction [S. Kaxiras and J. R. Goodman, ISCA 1999]
- Sharing prediction [A.-C. Lai and B. Falsafi, ISCA 1999]

Dead block elimination:

- Dynamic self-invalidation [A. R. Lebeck and D. A. Wood, ISCA 1995]
- Last touch prediction [A.-C. Lai and B. Falsafi, ISCA 2000]
- Dead block prediction [A.-C. Lai, C. Fide and B. Falsafi, ISCA 2001]

Stale data values:

- Coherence decoupling [J. Huh et al., ASPLOS 2004]
- Approximate stale loads [P. V. Rengasamy et al., PACT 2015]

Virtual Networks

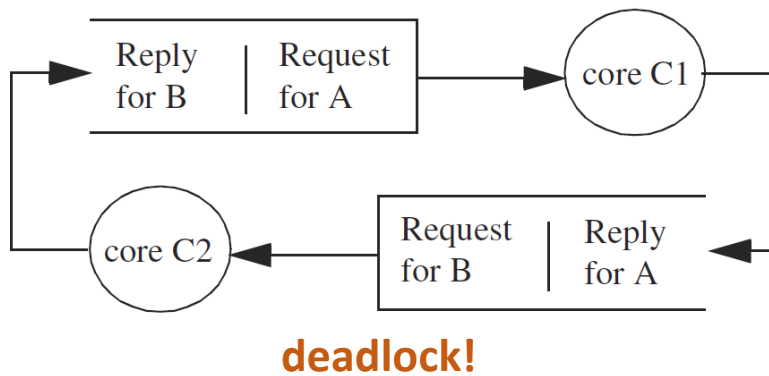
To avoid deadlock, dependent messages should not have to wait for each other. Use virtual channels (in network) and virtual queues (in cache/directory controllers) to logically isolate different message types.

- e.g., if message types X, Y and Z have dependence order $X \rightarrow Y \rightarrow Z$, all messages can share the same physical channels and queues, as long as Z messages can bypass X and Y, and Y can bypass X.

Virtual Networks

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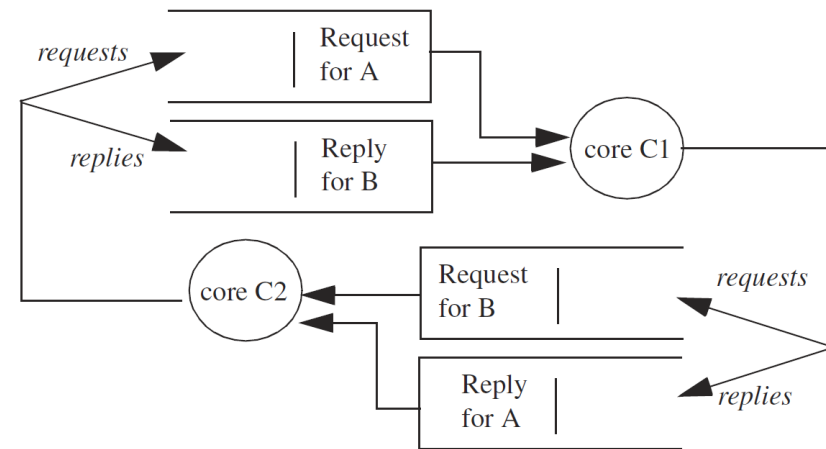
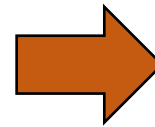
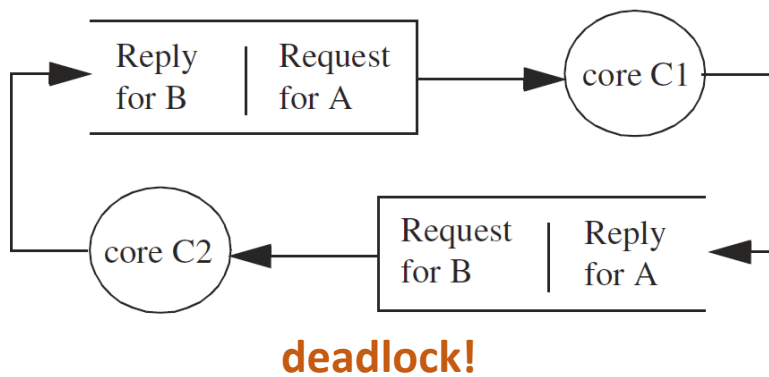
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Virtual Networks

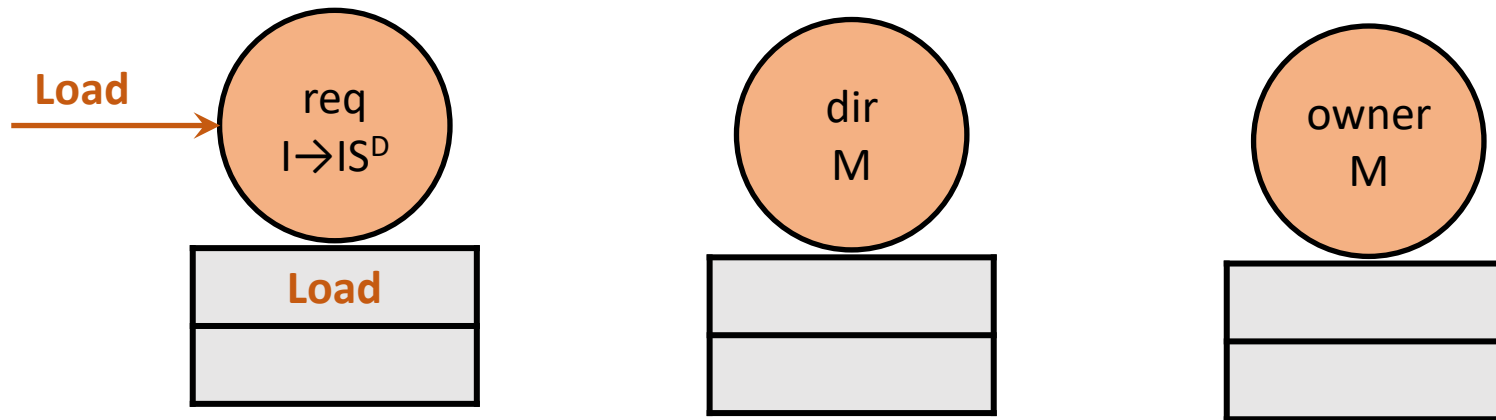
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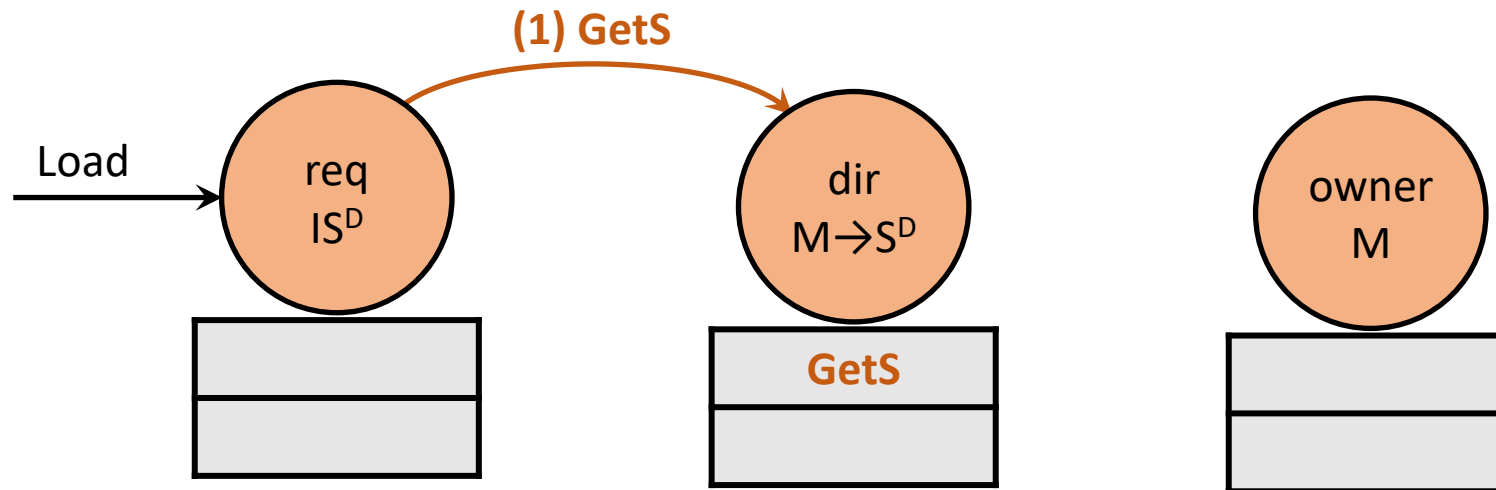
Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol



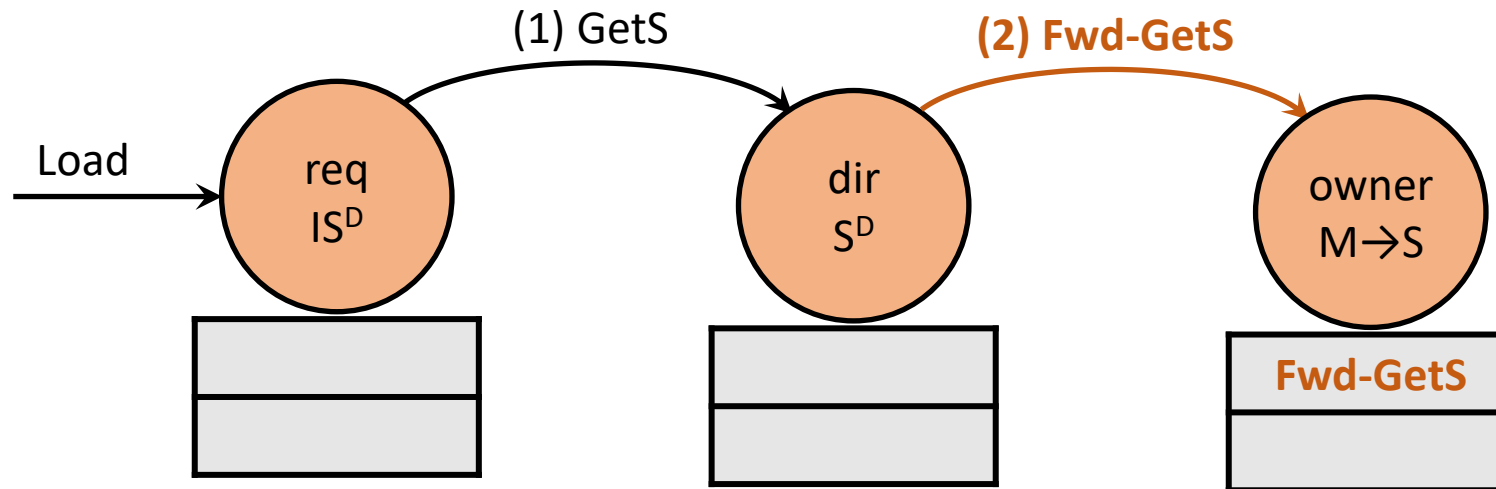
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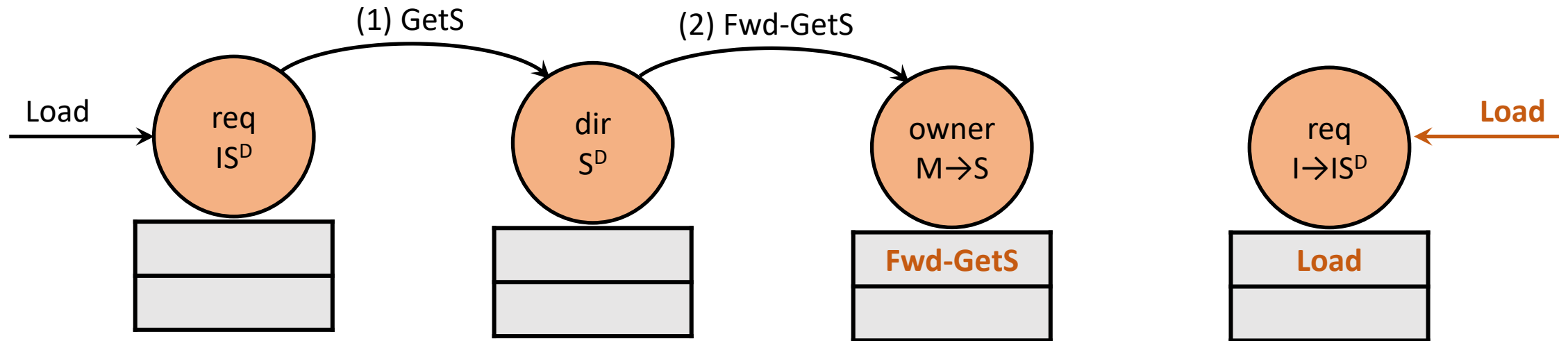
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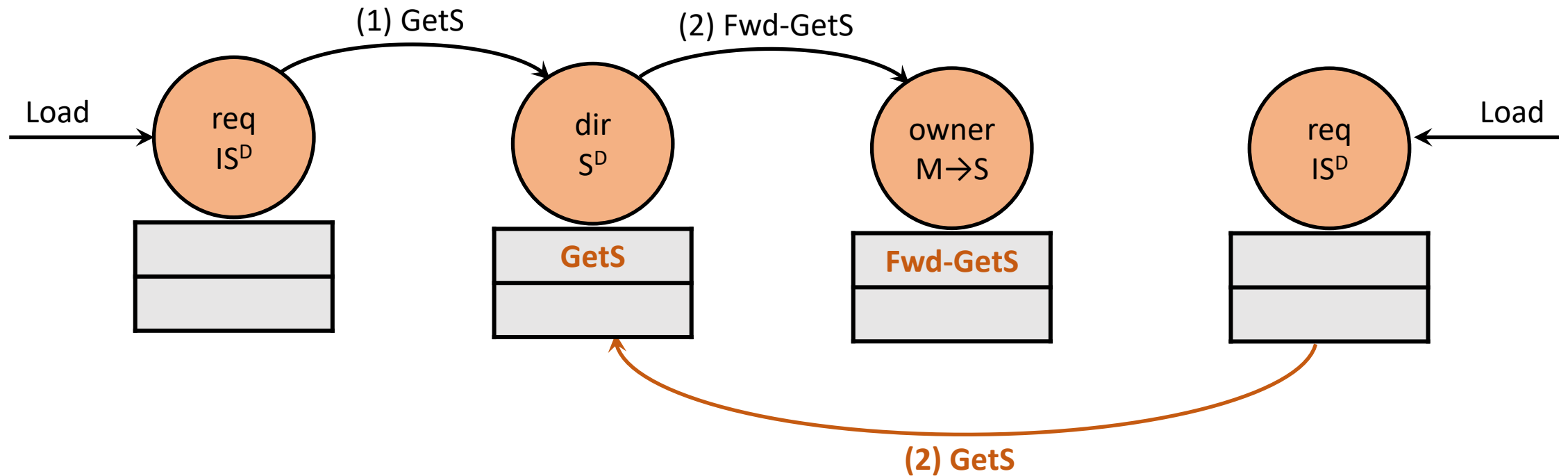
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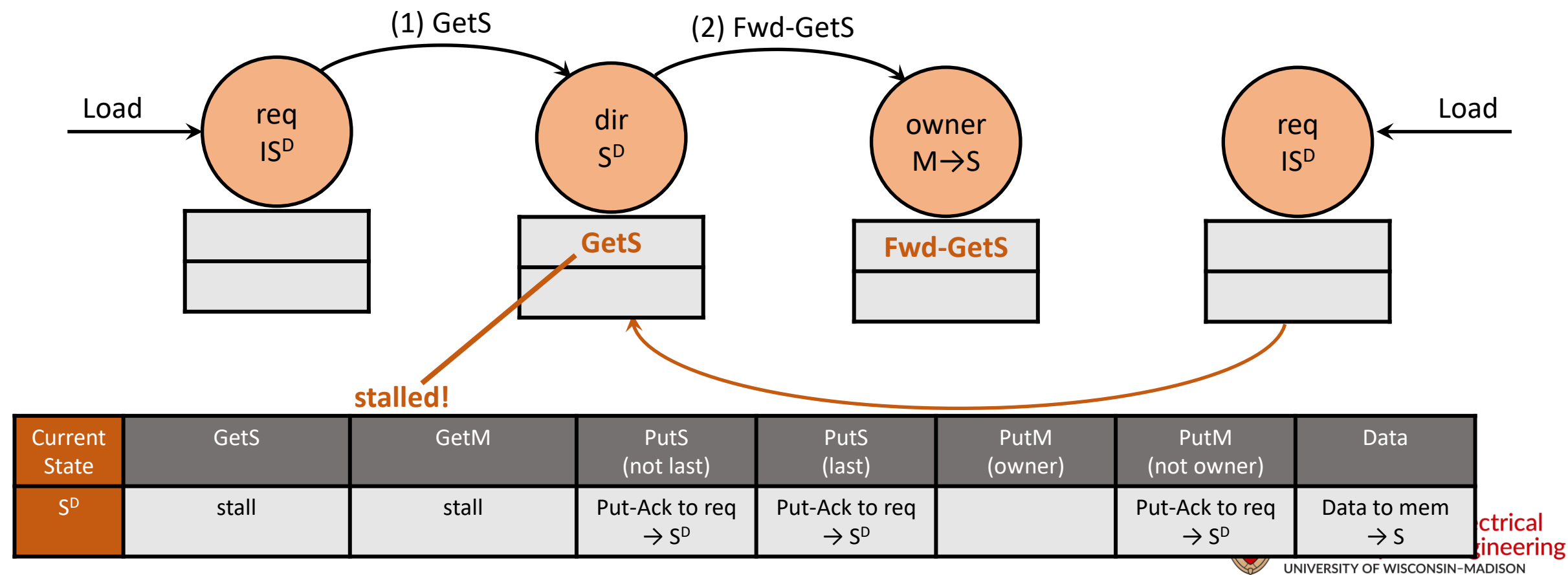
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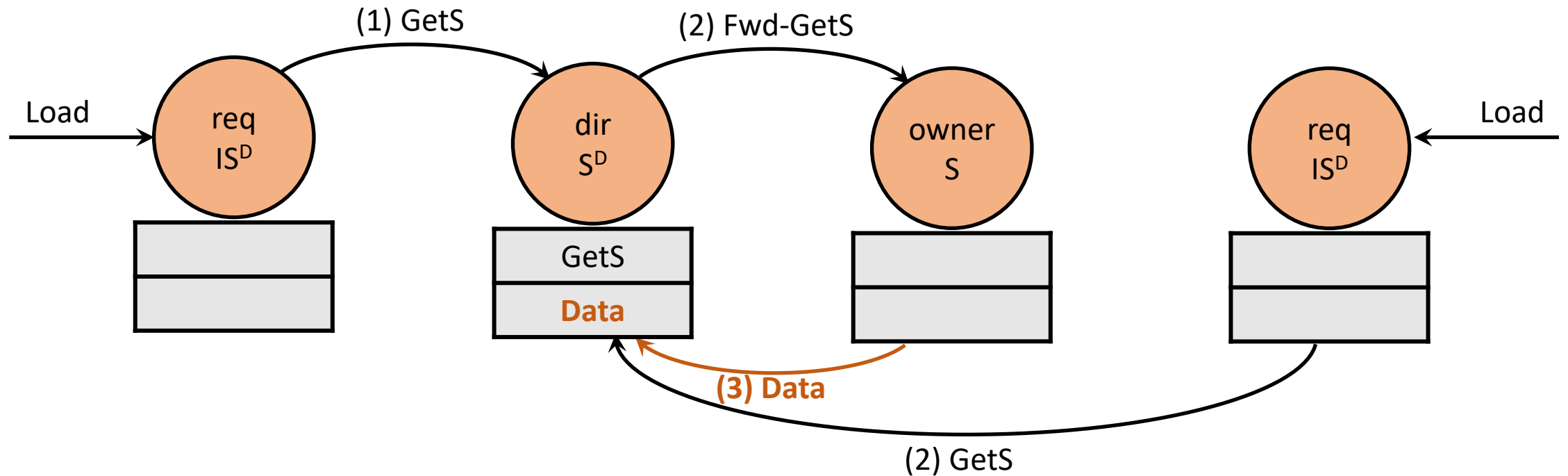
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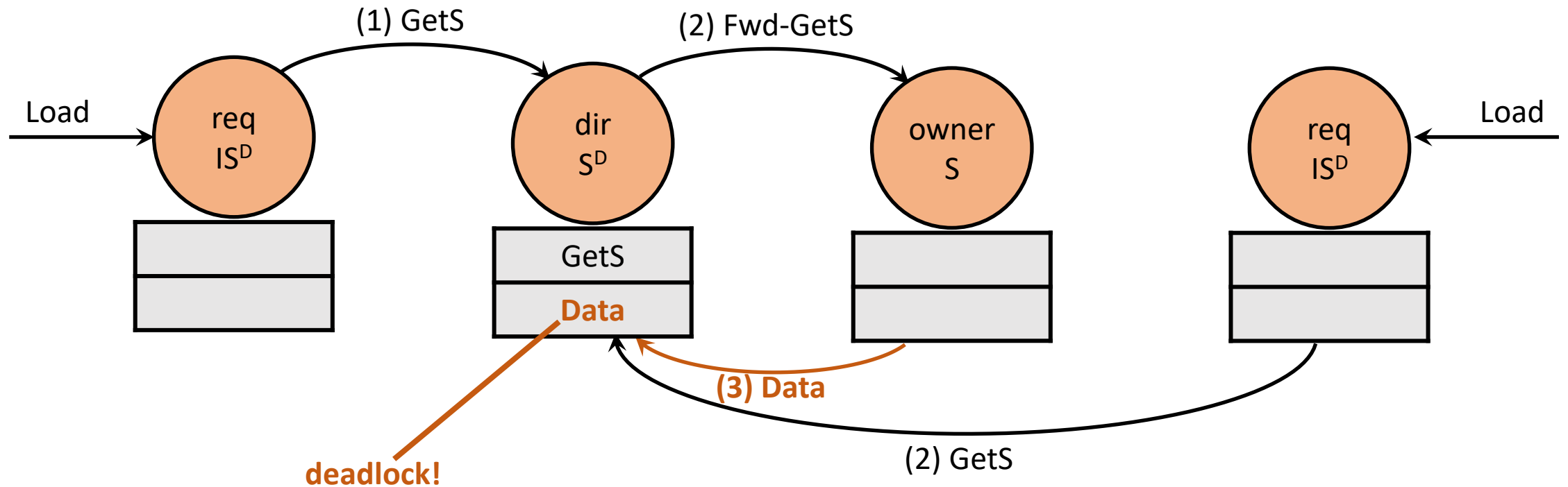
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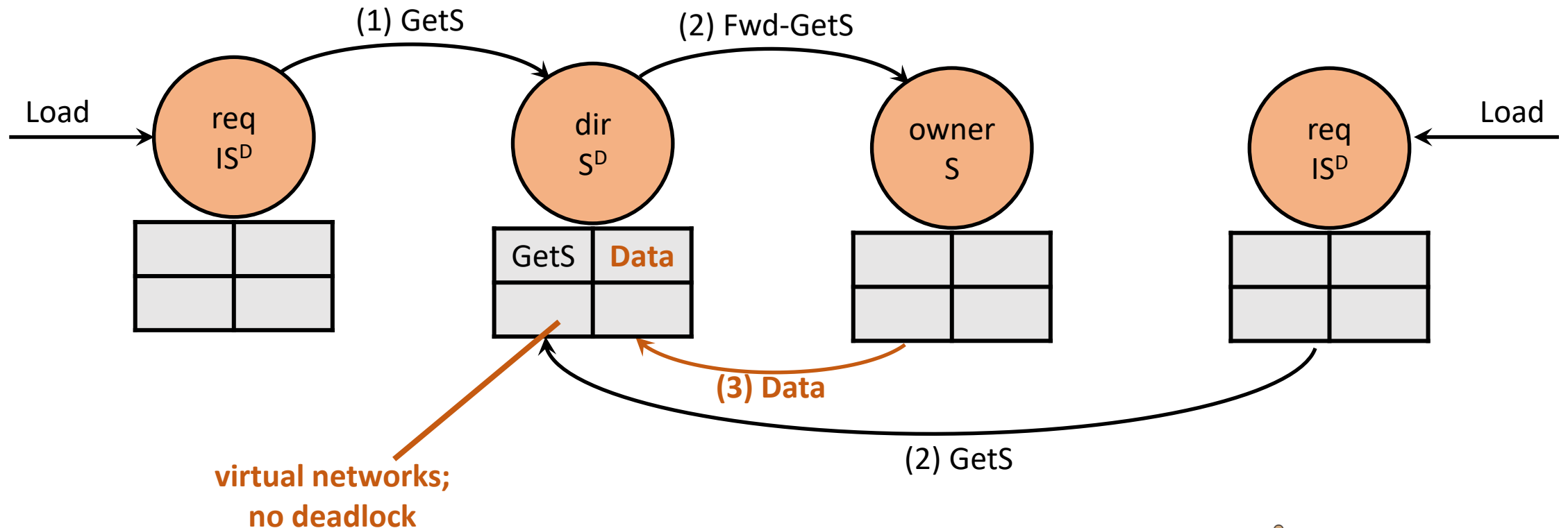
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Virtual Networks – MSI

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Virtual Networks – MSI

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An analogy:

Cars stopped on road while waiting to pick up bus passengers, but blocking bus.

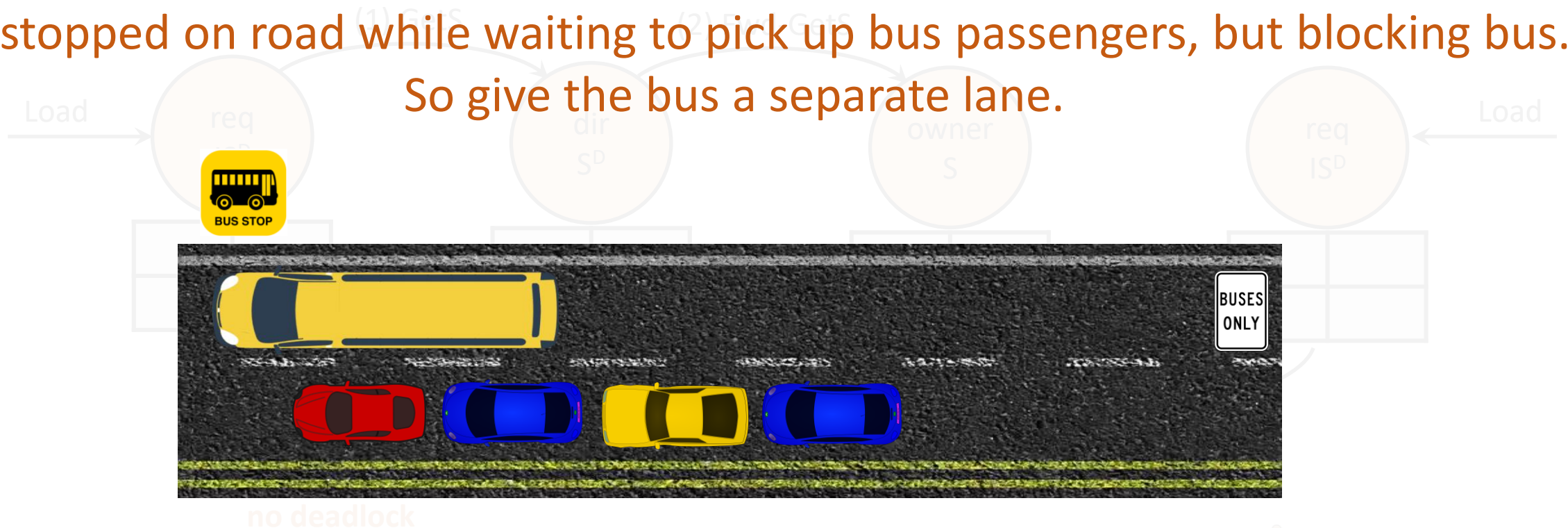


Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

An analogy:

Cars stopped on road while waiting to pick up bus passengers, but blocking bus.
So give the bus a separate lane.



Virtual Networks – MSI

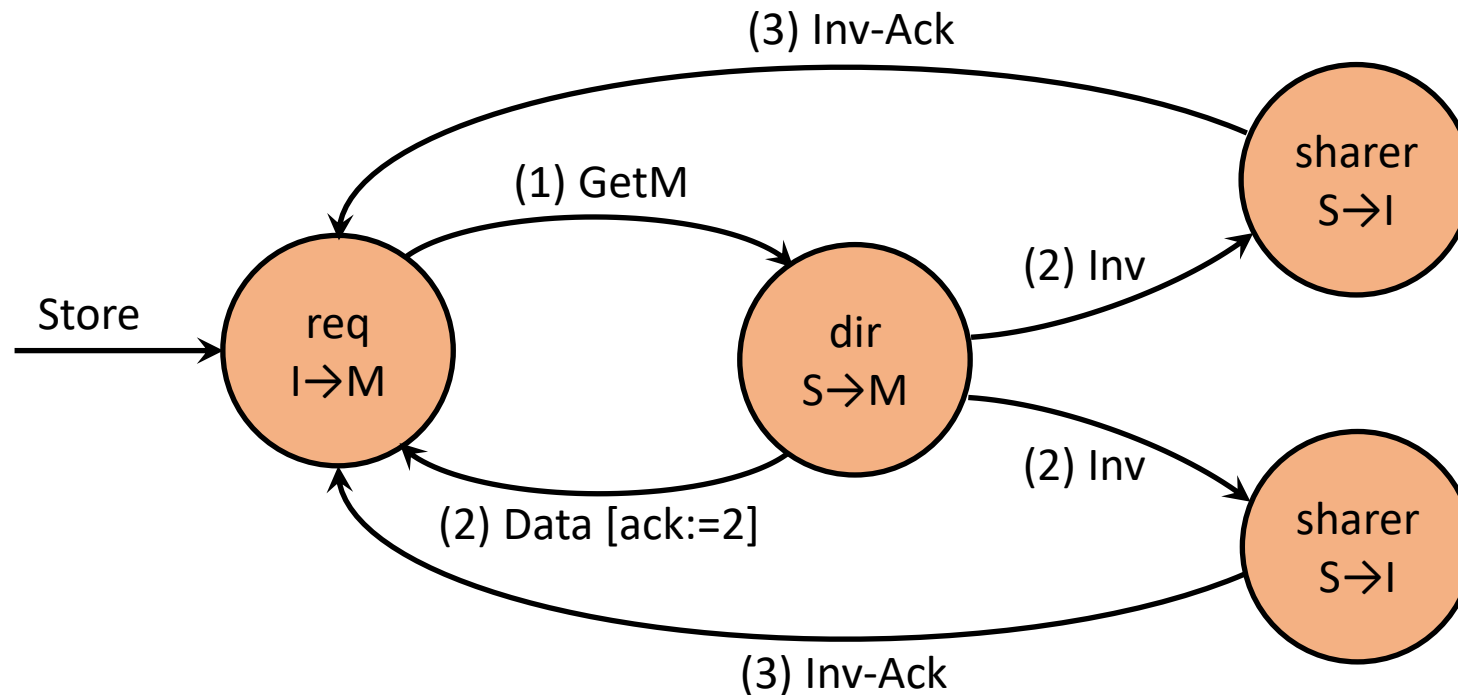
Modified-Shared-Invalid (MSI) Protocol

- How many virtual networks to prevent deadlock?

Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

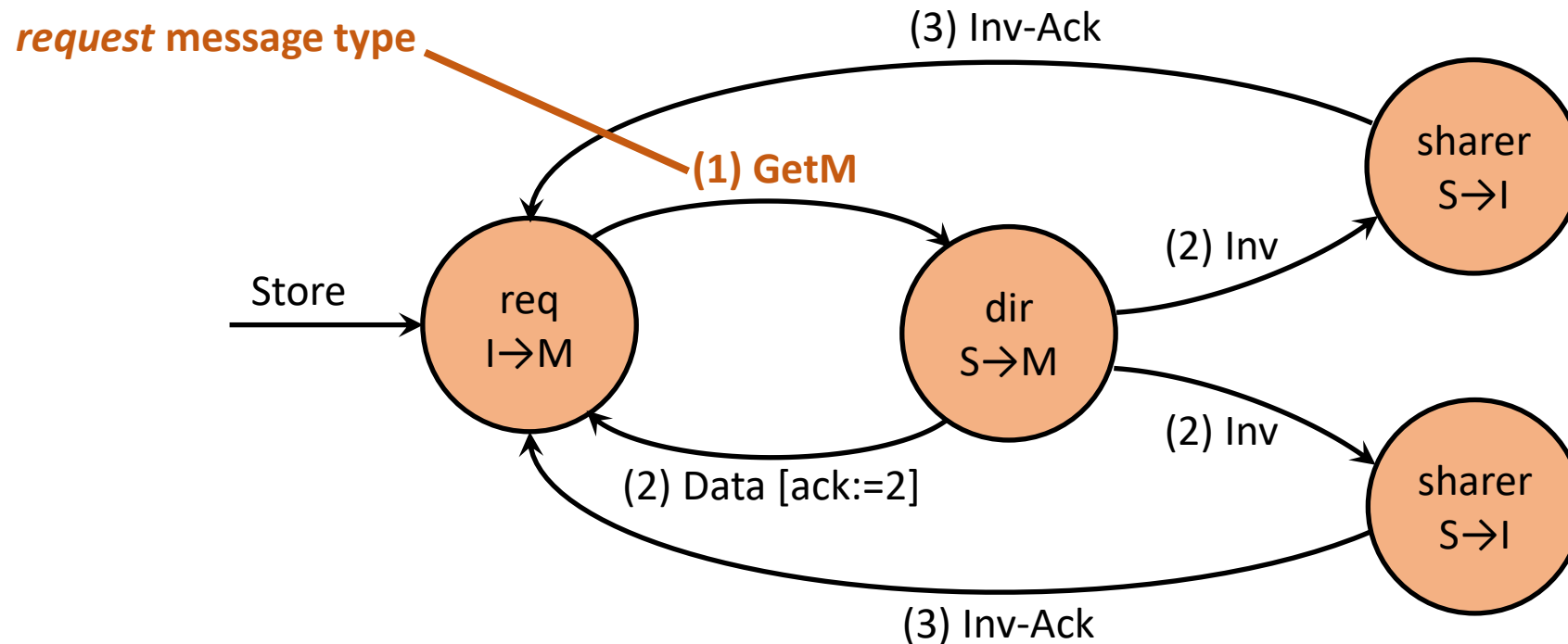
- How many virtual networks to prevent deadlock?



Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

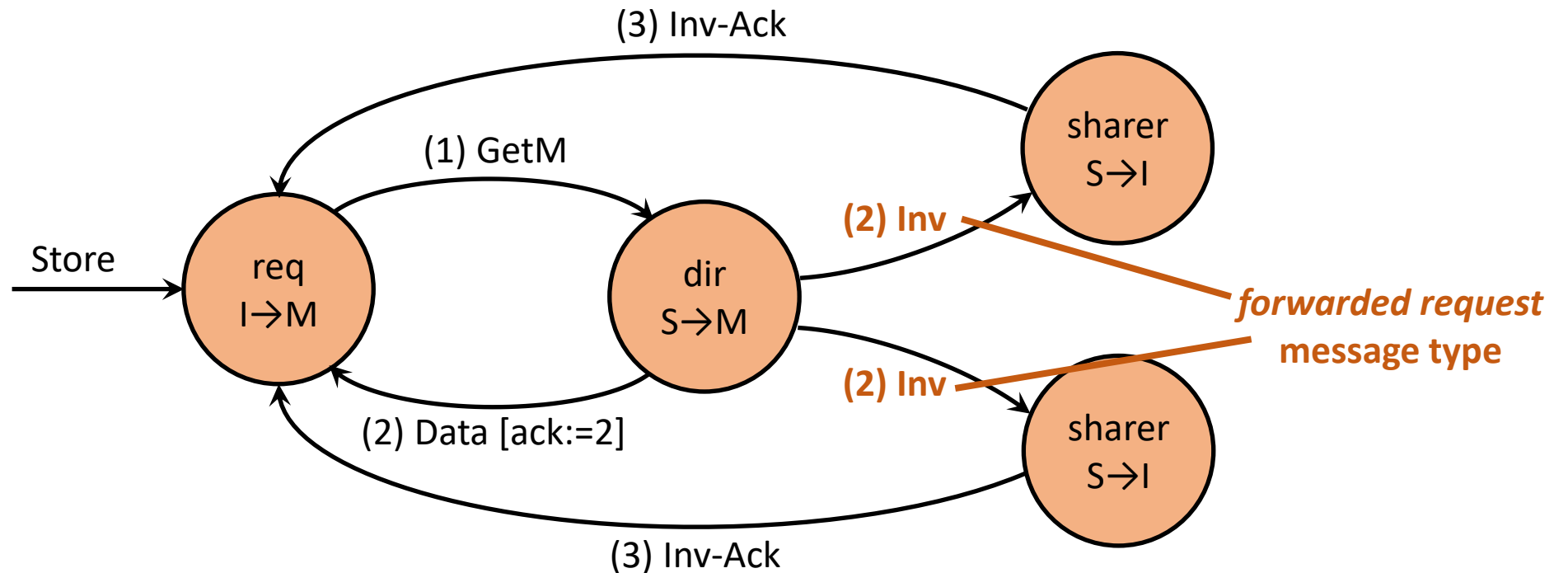
- How many virtual networks to prevent deadlock?



Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

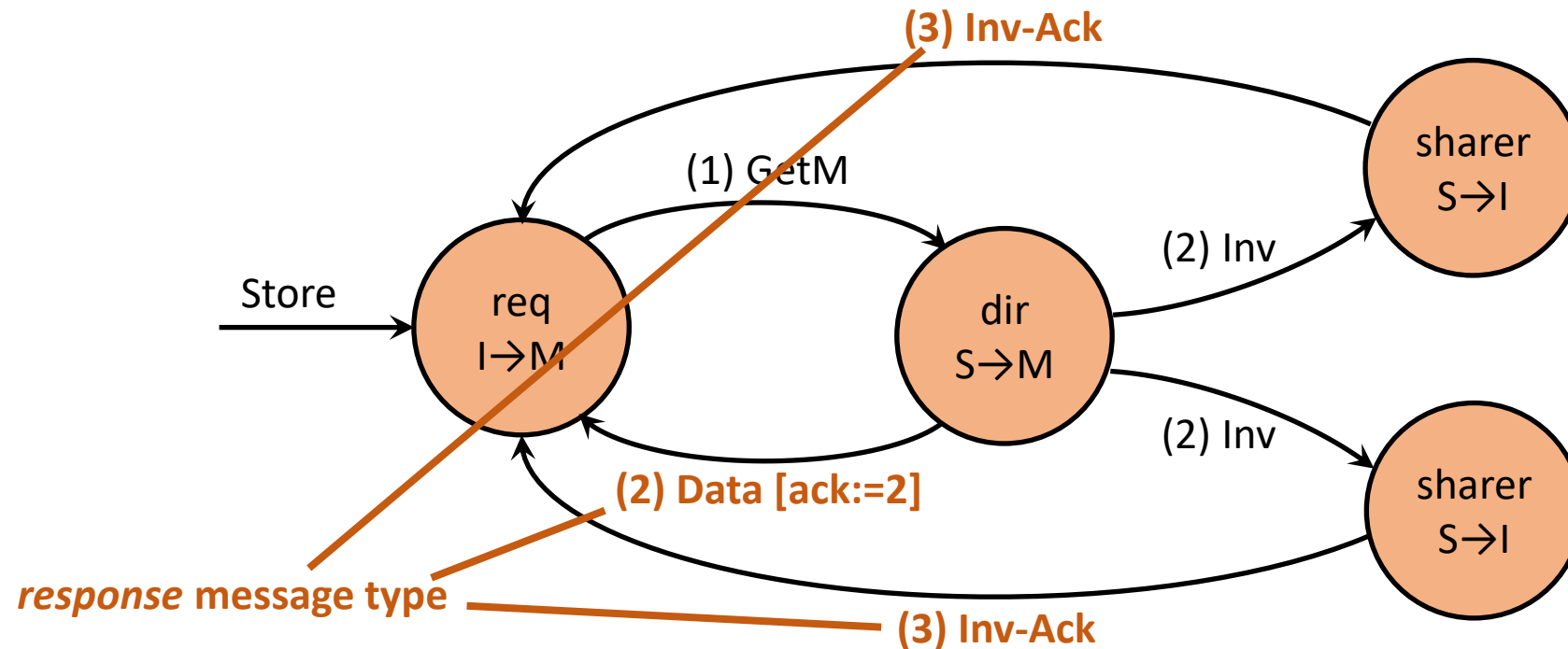
- How many virtual networks to prevent deadlock?



Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

- How many virtual networks to prevent deadlock?



Virtual Networks – MSI

Modified-Shared-Invalid (MSI) Protocol

- How many virtual networks to prevent deadlock?

Request messages: GetS, GetM, PutS, PutM

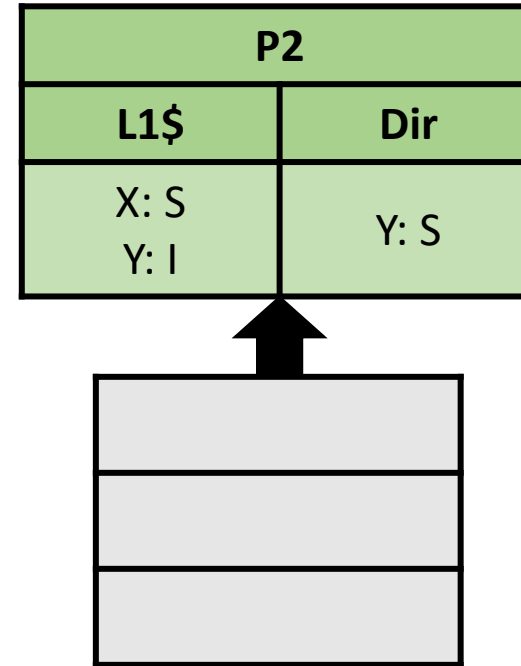
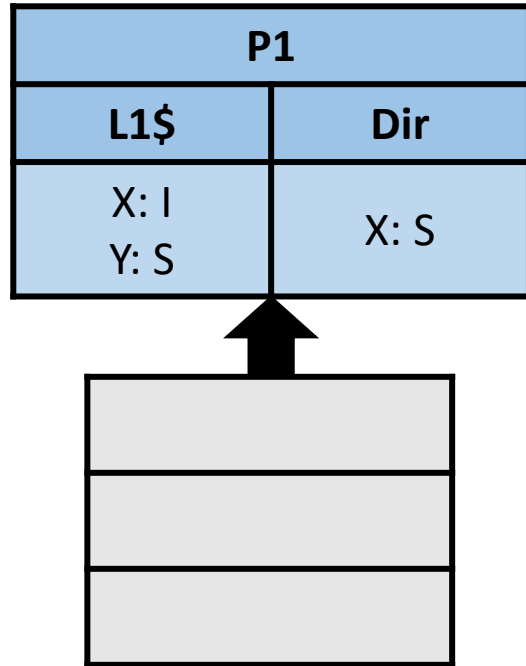
Forwarded request messages: Fwd-GetS, Fwd-GetM, Inv

Response messages: Inv-Ack, Put-Ack, Data

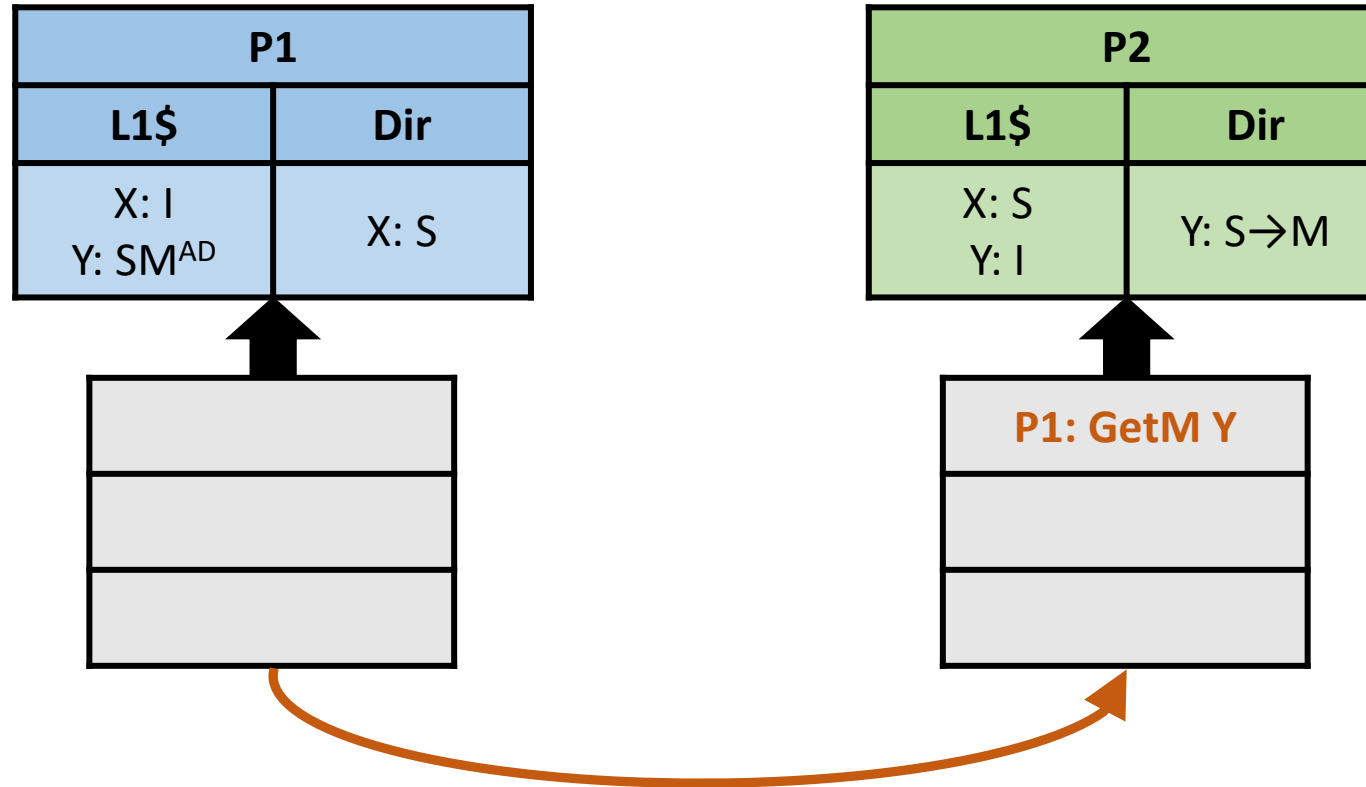
...at least 3 virtual networks needed*

***Assuming all nodes (i.e., private caches, directory banks) share the same physical network**

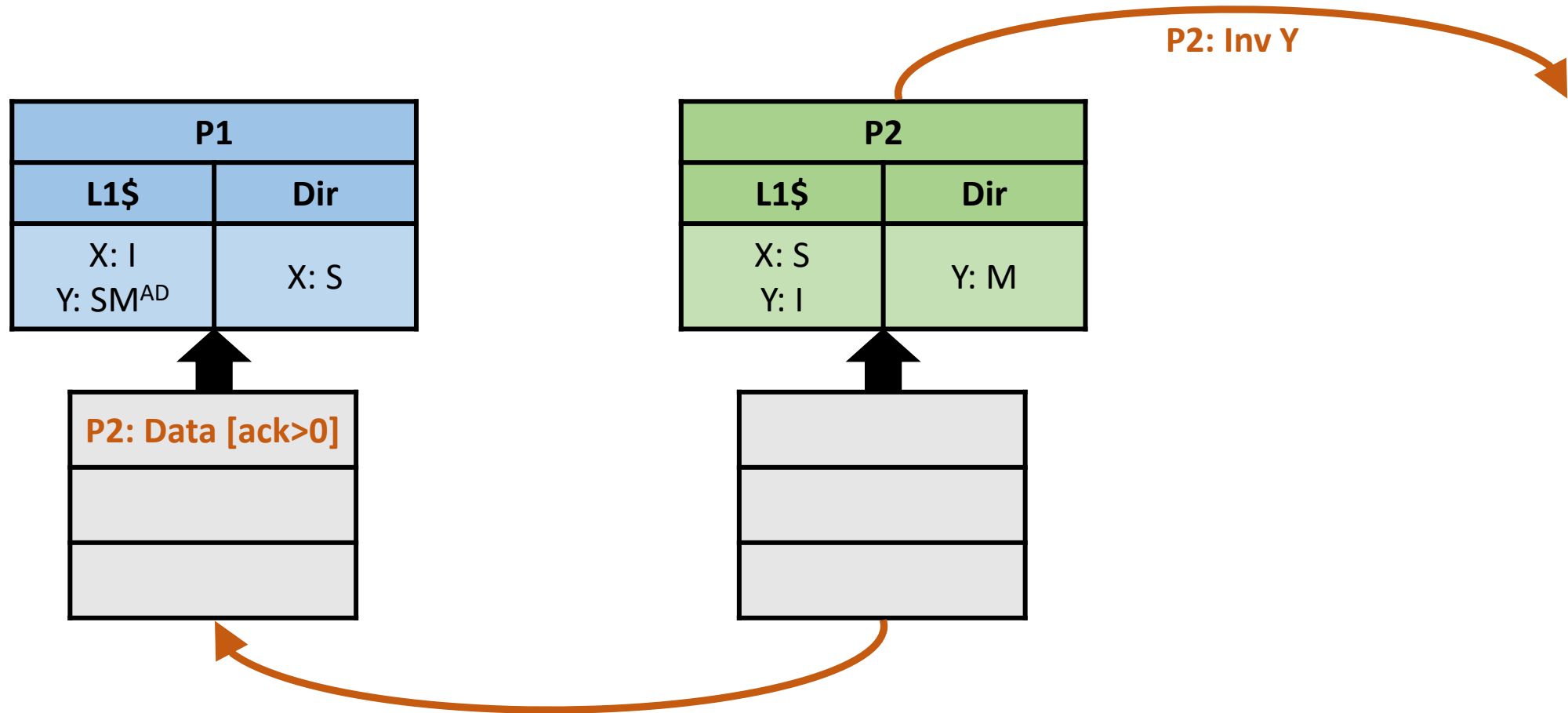
Virtual Networks – MSI



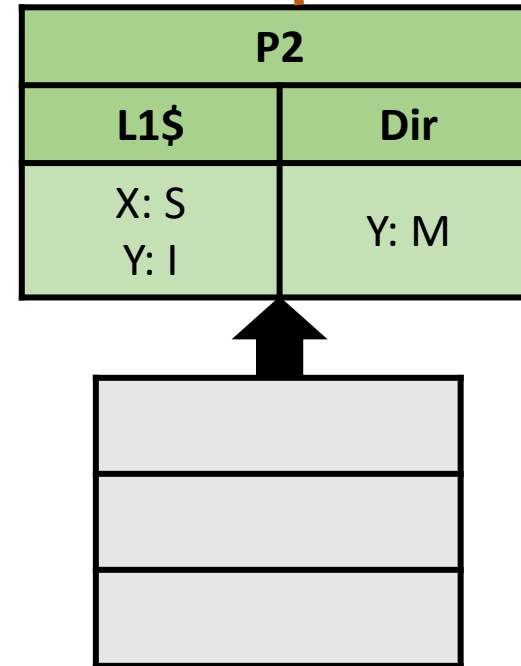
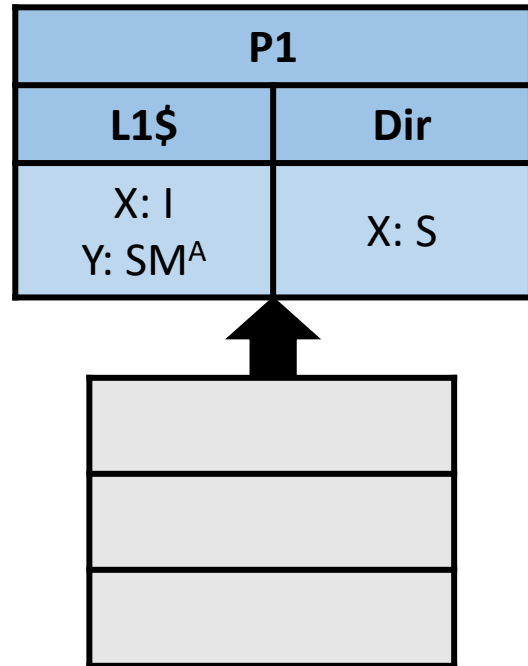
Virtual Networks – MSI



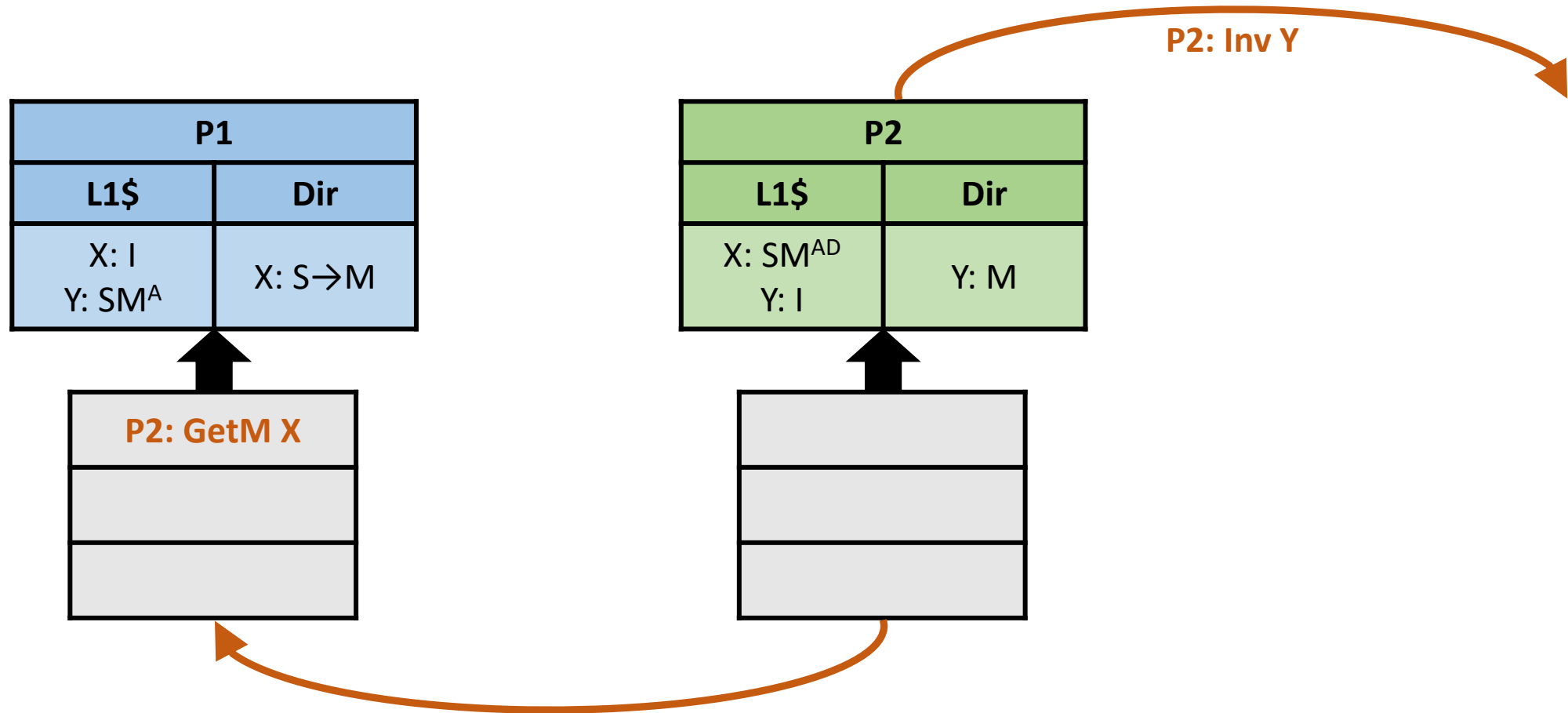
Virtual Networks – MSI



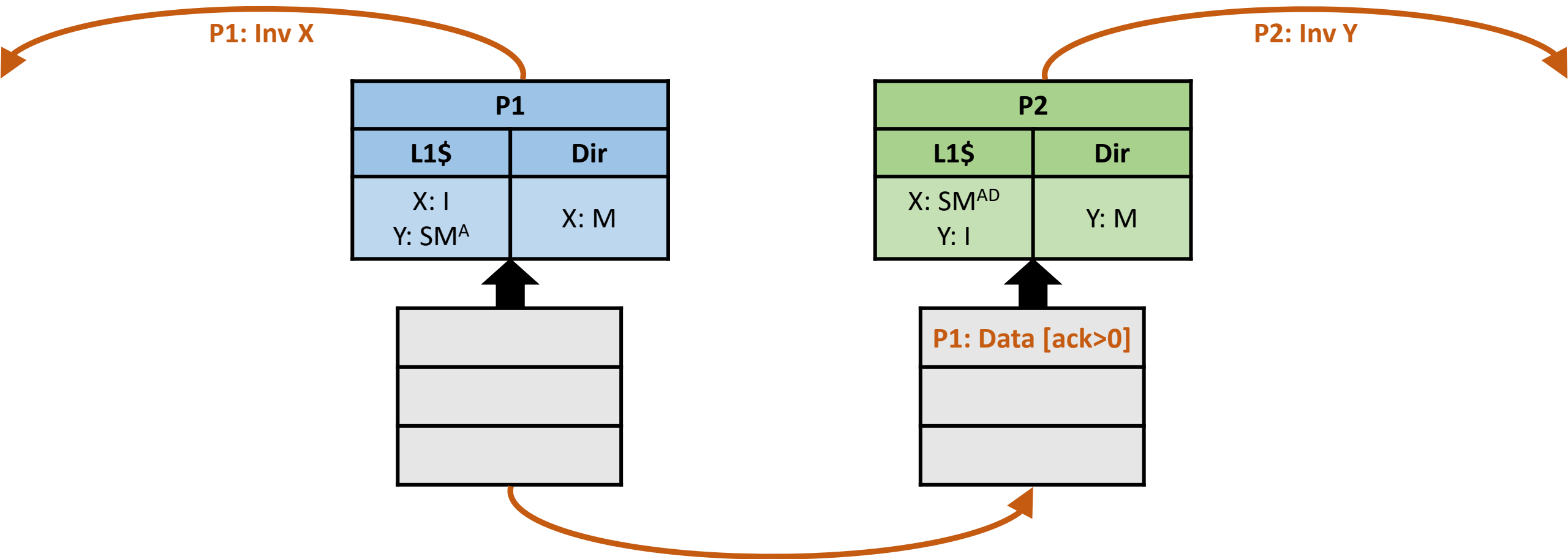
Virtual Networks – MSI



Virtual Networks – MSI

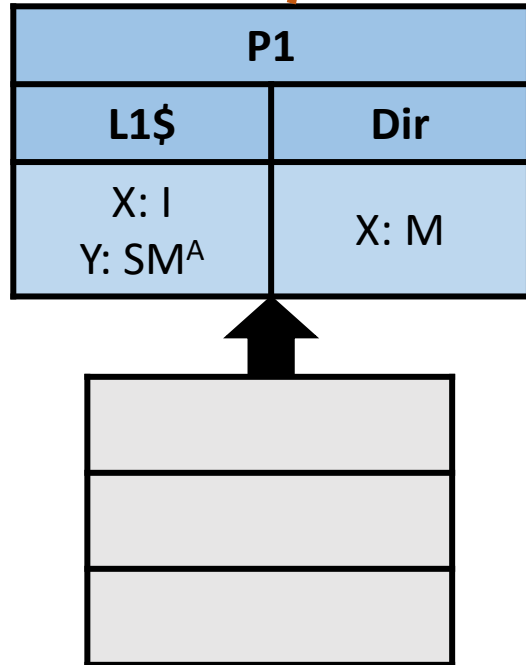


Virtual Networks – MSI

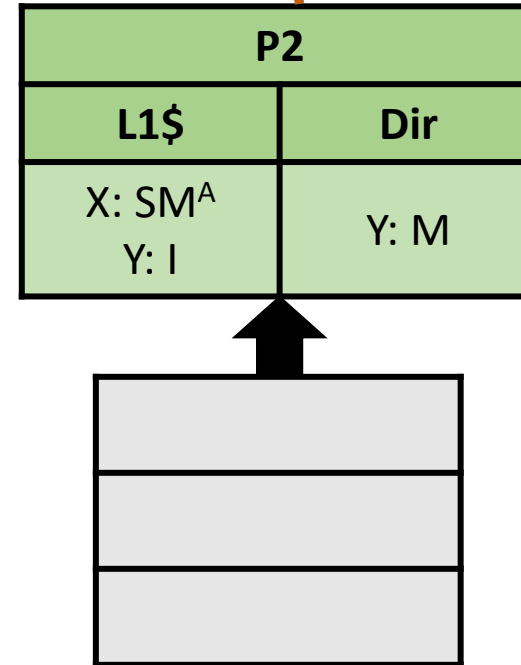


Virtual Networks – MSI

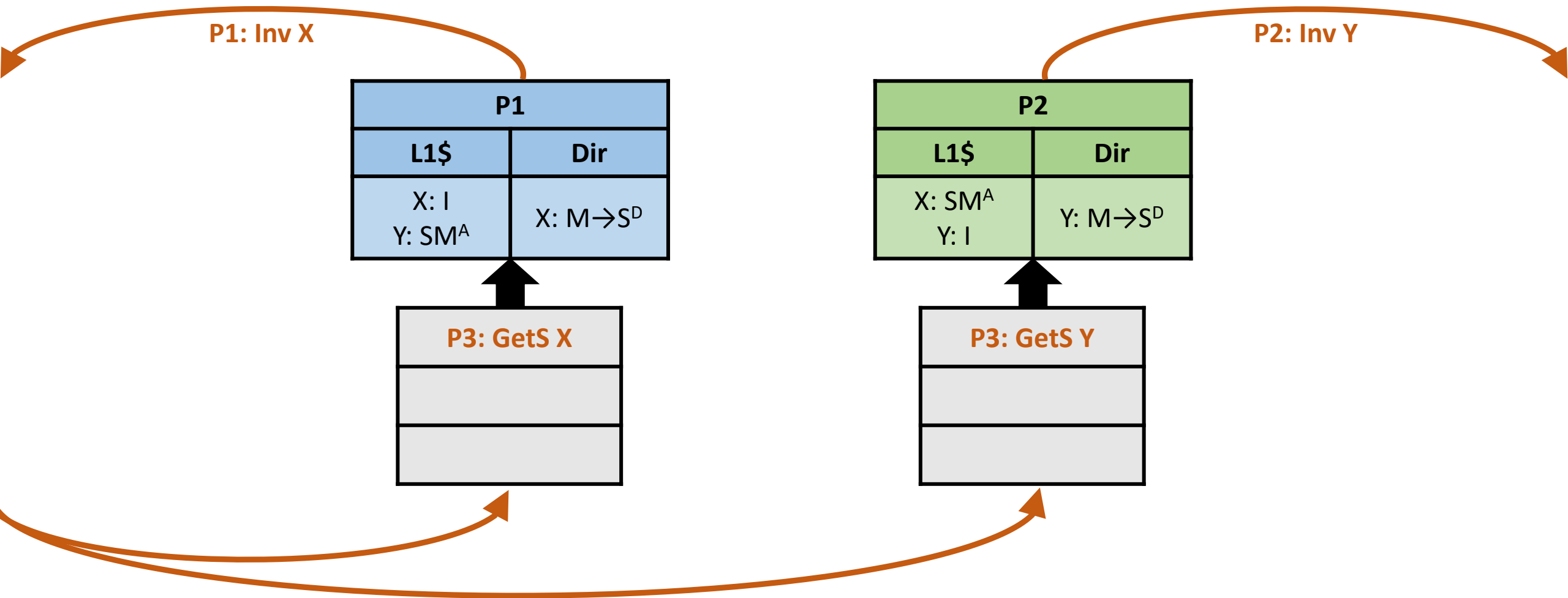
P1: Inv X



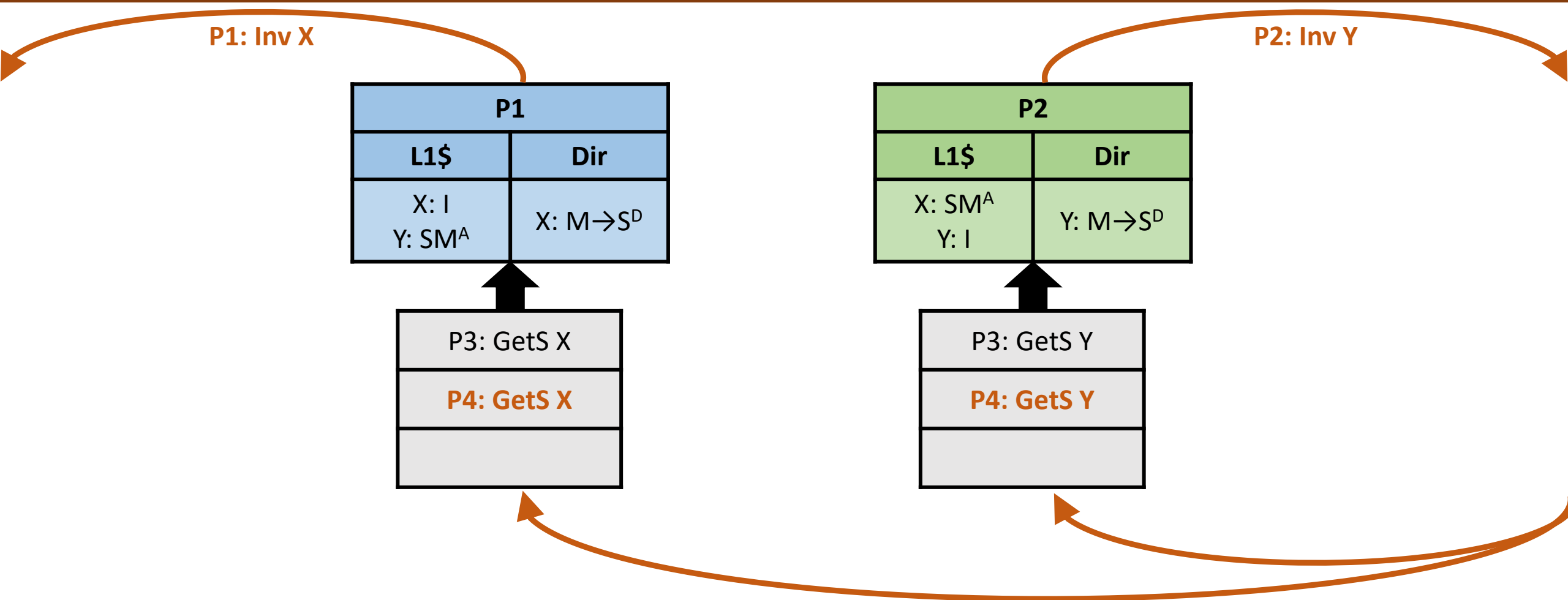
P2: Inv Y



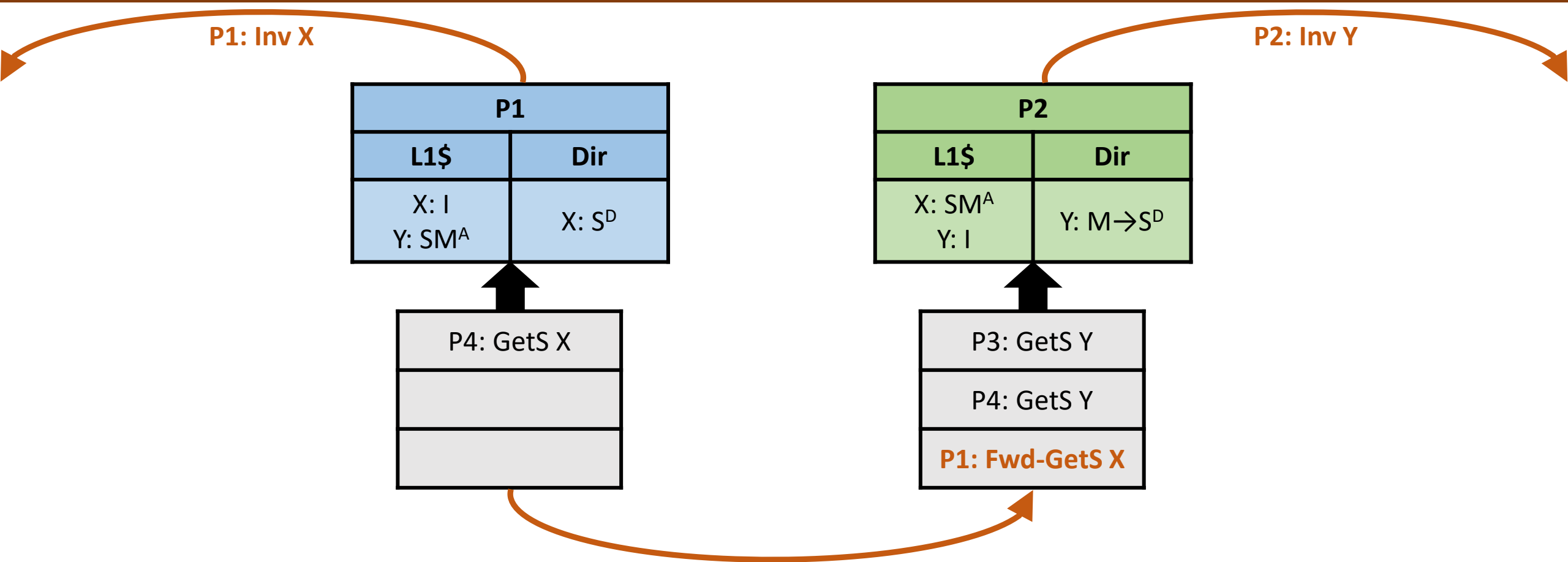
Virtual Networks – MSI



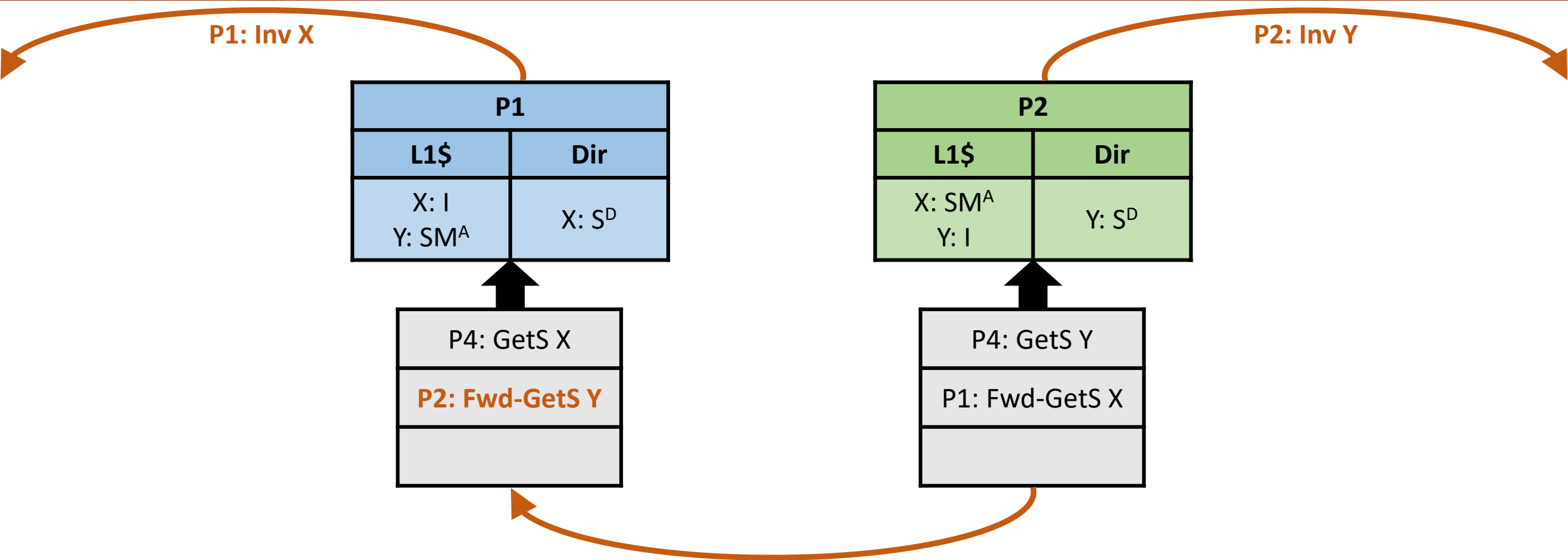
Virtual Networks – MSI



Virtual Networks – MSI



Virtual Networks – MSI



Virtual Networks – MSI

P1: Inv X

P1	
L1\$	Dir
X: I Y: SM ^A	X: S ^D

P4: GetS X
P2: Fwd-GetS Y

P2: Inv Y

P2	
L1\$	Dir
X: SM ^A Y: I	Y: S ^D

P4: GetS Y
P1: Fwd-GetS X



Virtual Networks – MSI

P1	
L1\$	Dir
X: I Y: SM ^A	X: S ^D

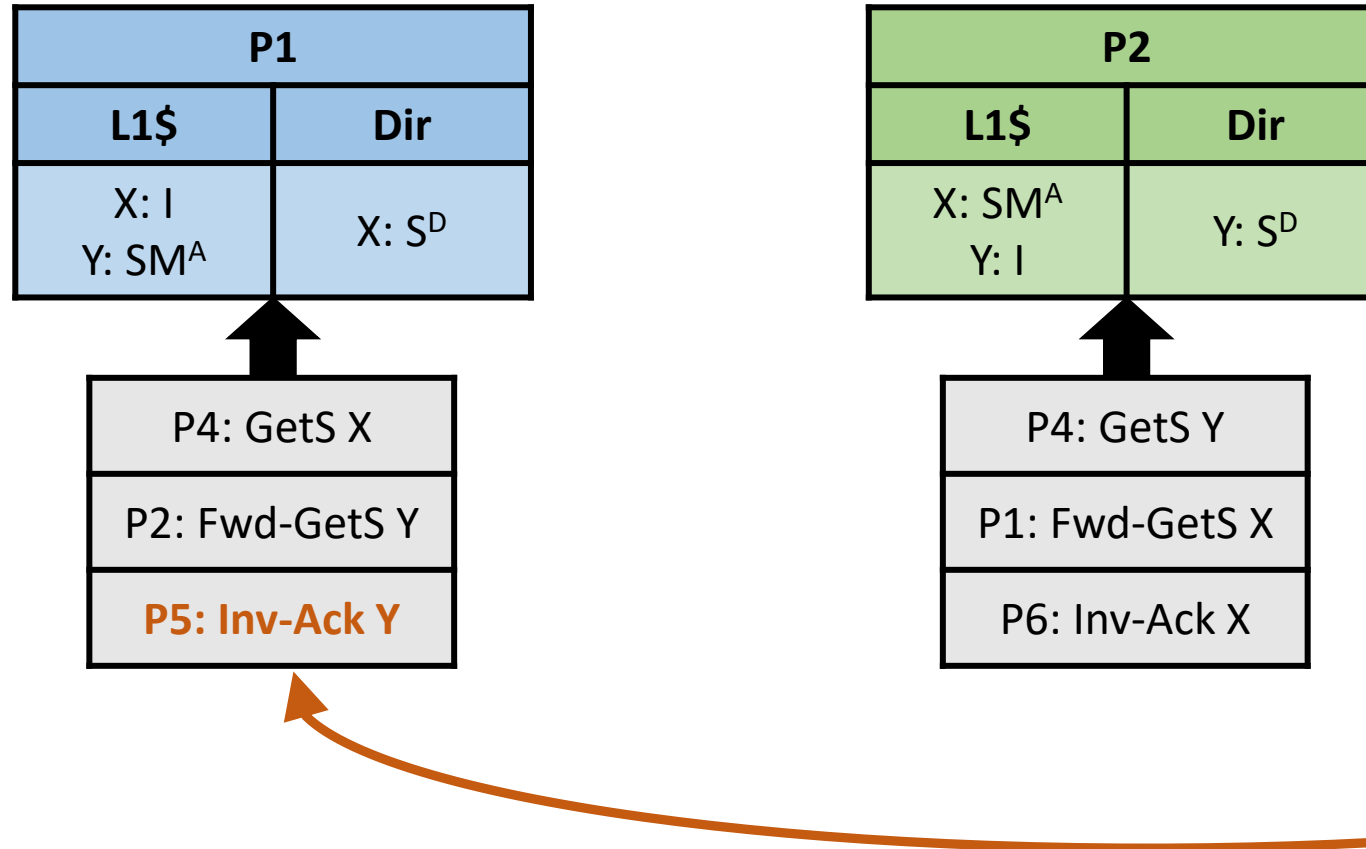
P4: GetS X
P2: Fwd-GetS Y

P2	
L1\$	Dir
X: SM ^A Y: I	Y: S ^D

P4: GetS Y
P1: Fwd-GetS X
P6: Inv-Ack X

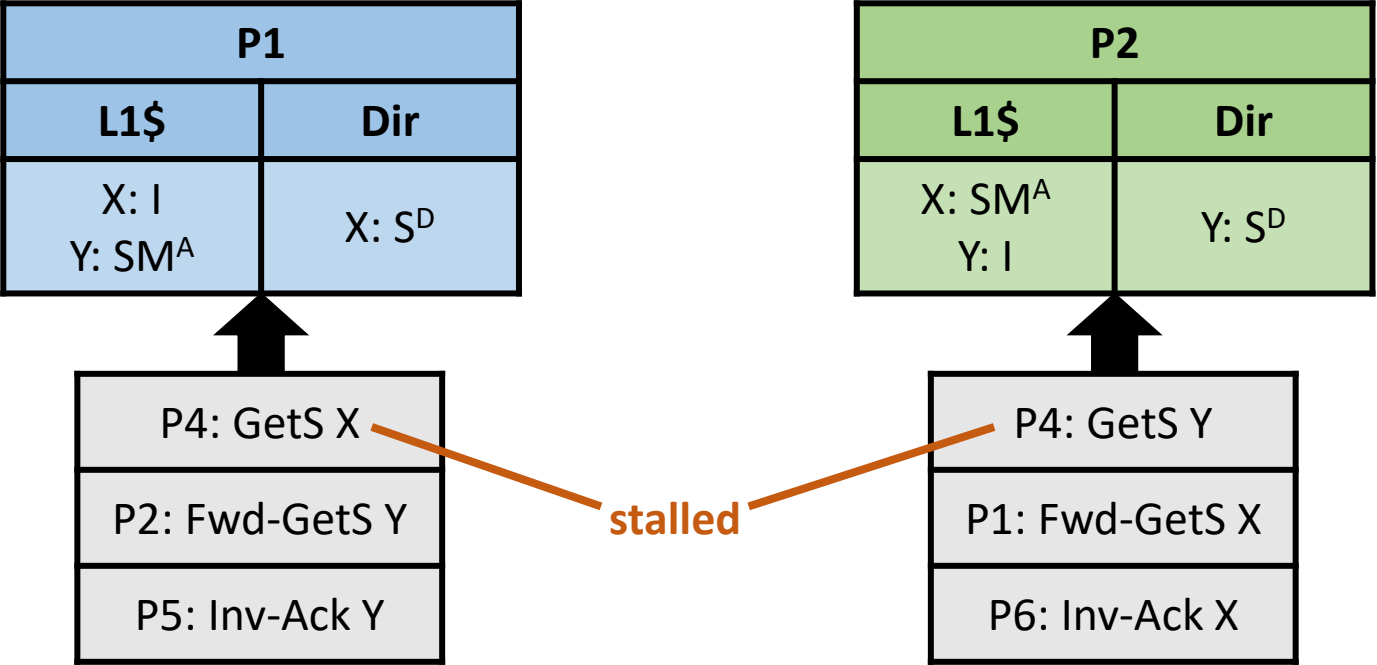
P2: Inv Y

Virtual Networks – MSI



Virtual Networks – MSI

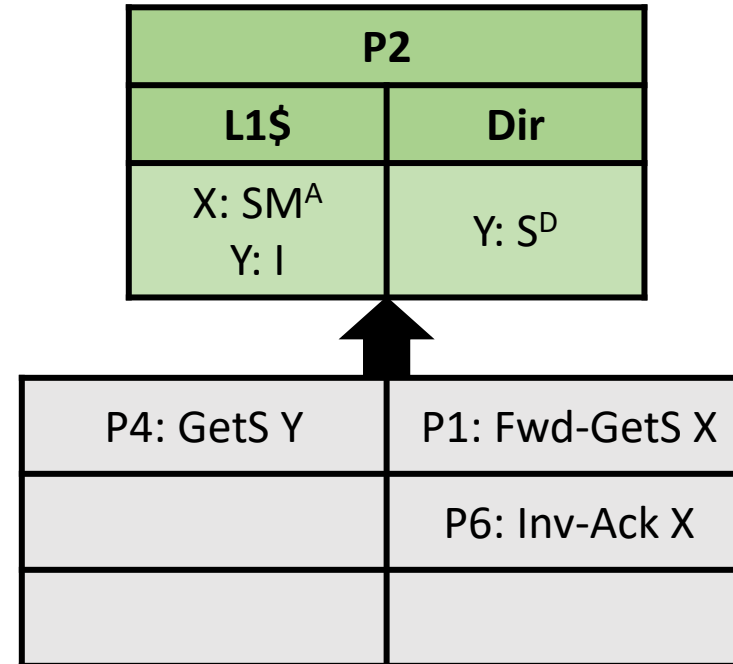
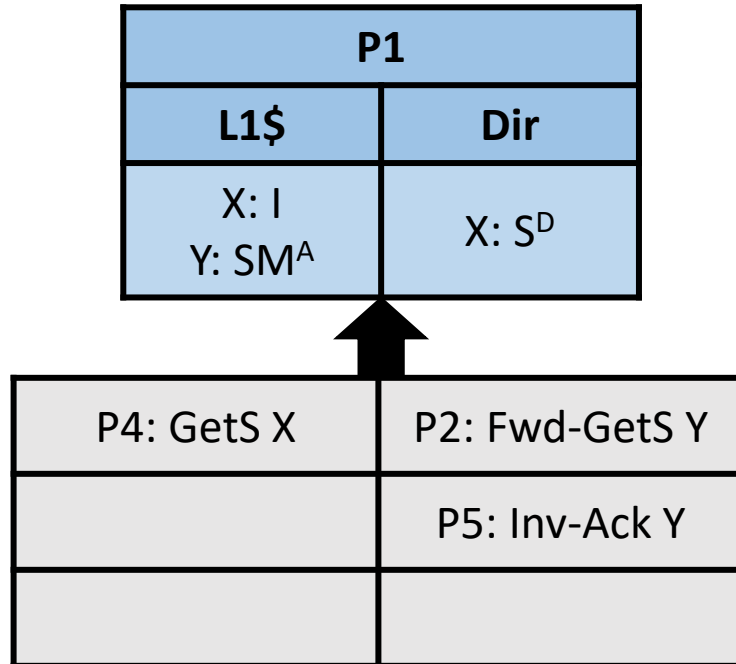
deadlock!



Directory Controller

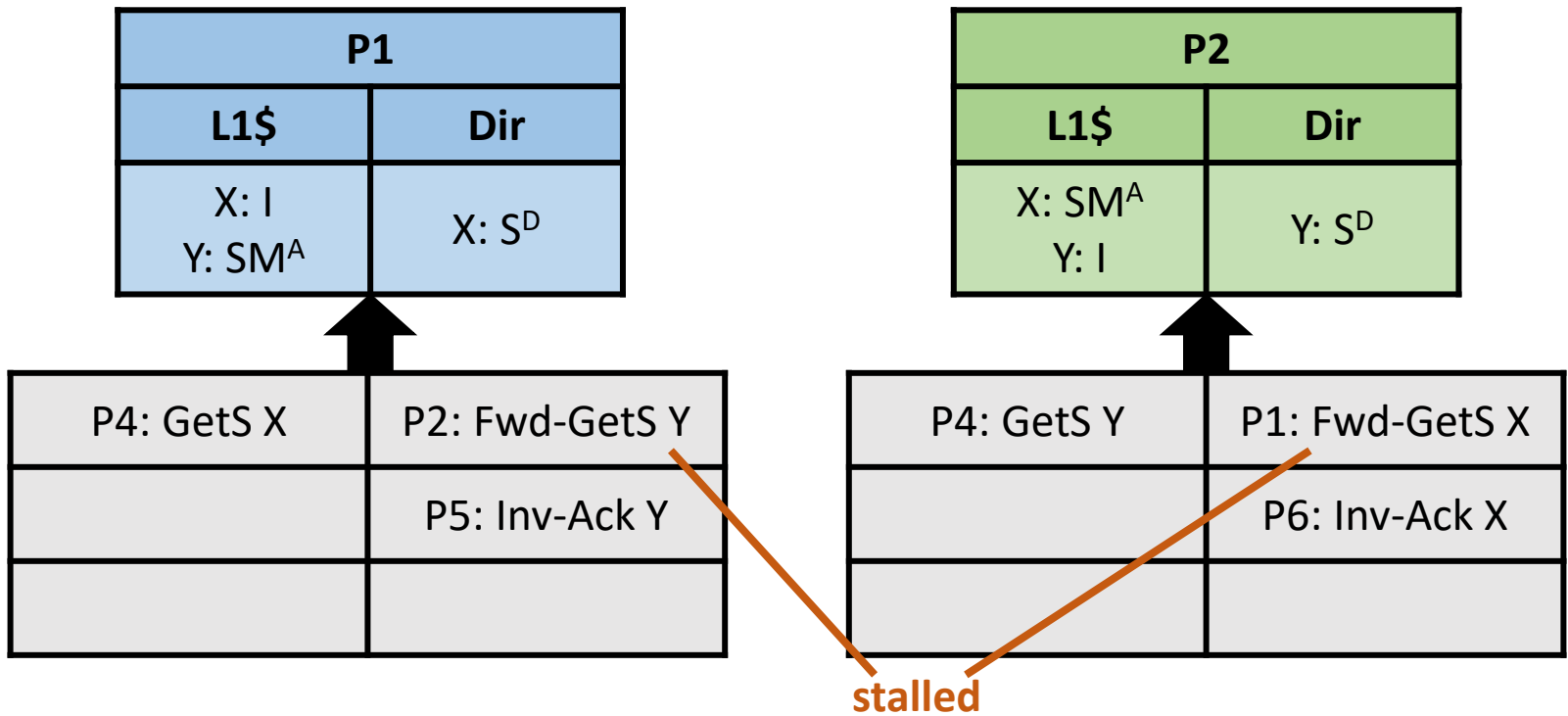
Current State	Event						
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
S^D	stall	stall	Put-Ack to req $\rightarrow S^D$	Put-Ack to req $\rightarrow S^D$		Put-Ack to req $\rightarrow S^D$	Data to mem $\rightarrow S$

Virtual Networks – MSI



Virtual Networks – MSI

deadlock!



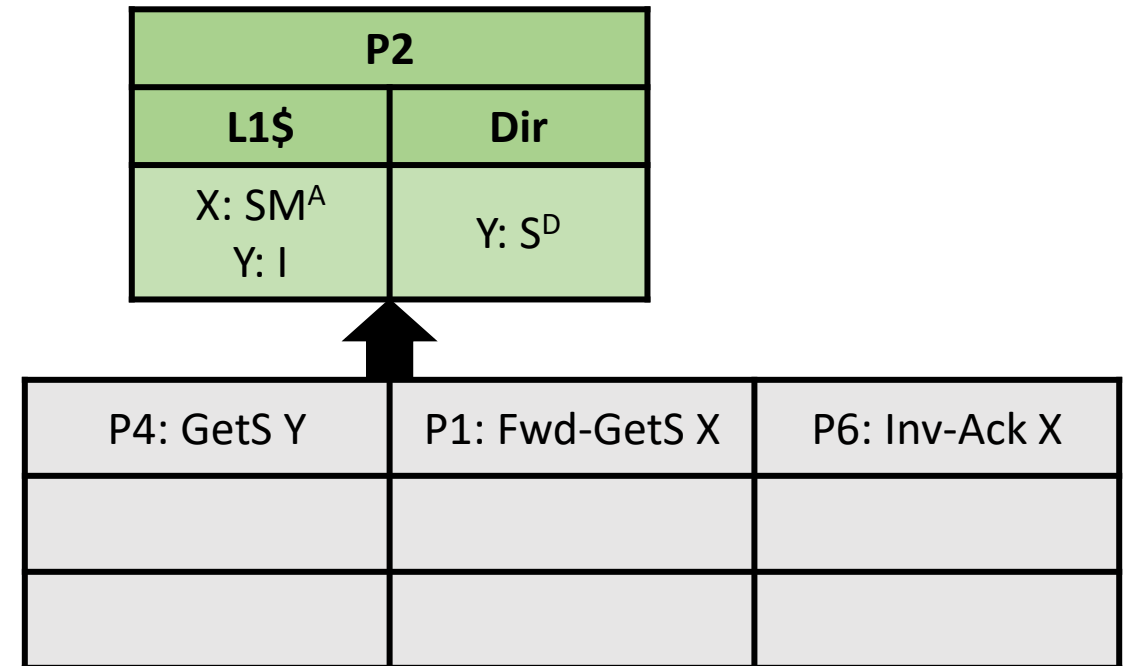
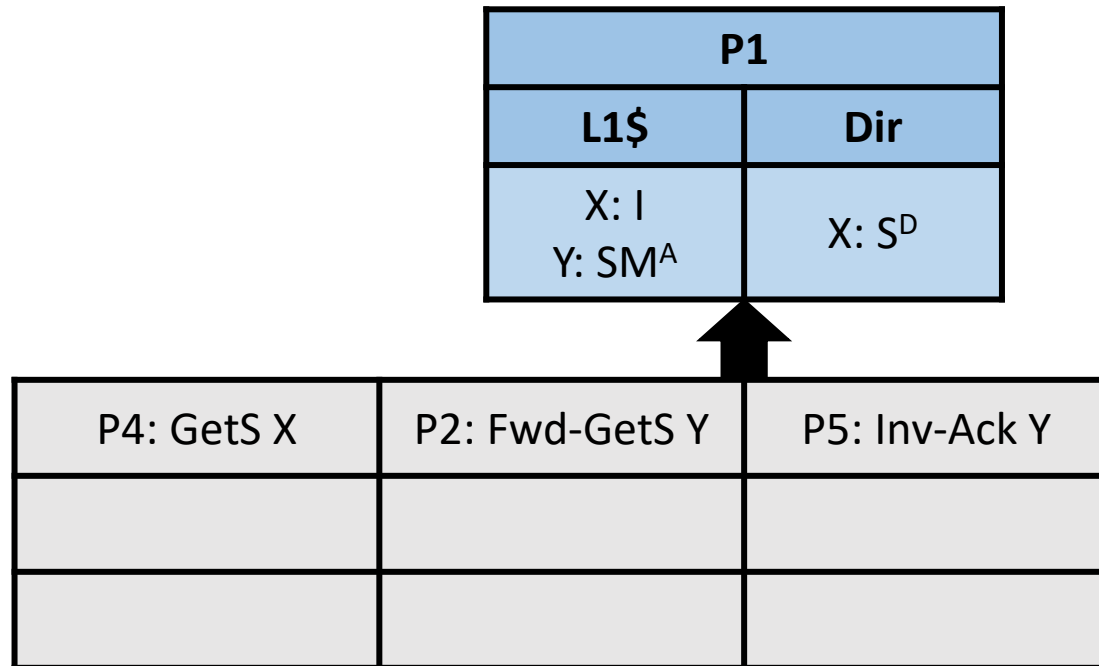
stalled

Cache Controller

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
SM ^A	hit → SM ^A	stall	stall	stall	stall						ack-- → SM ^A	→ M

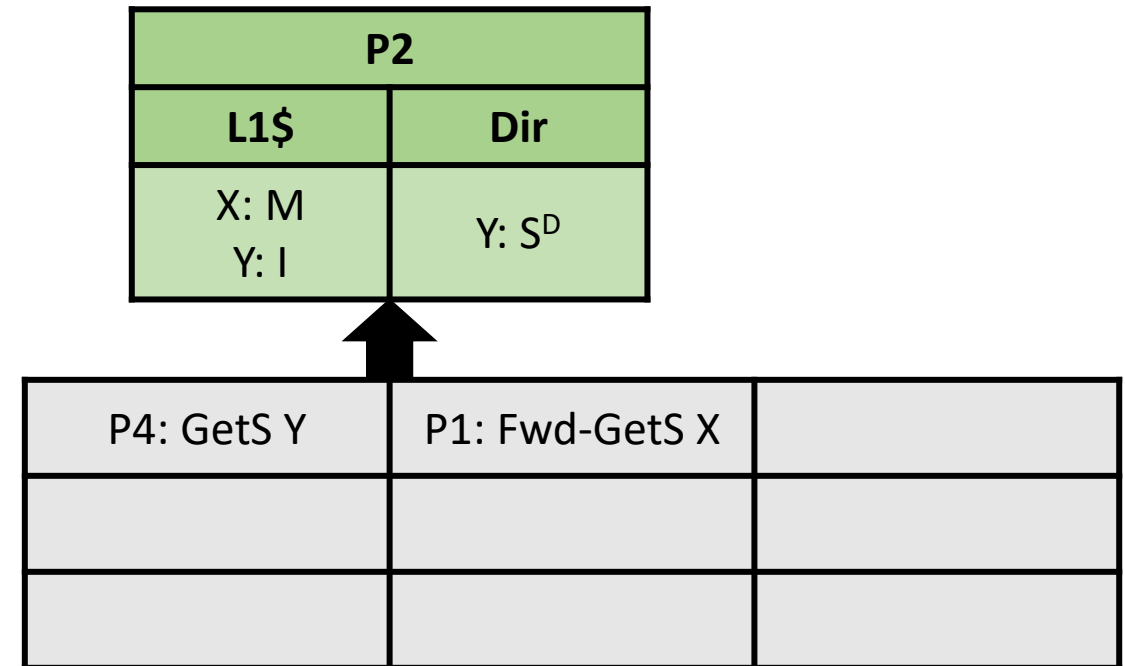
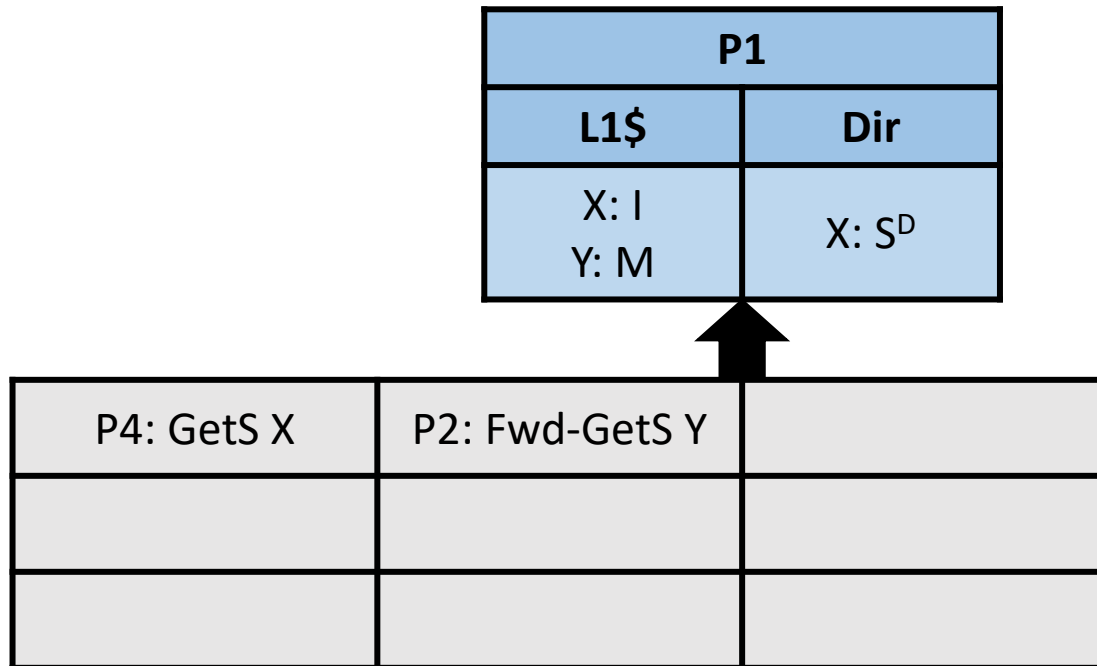
Virtual Networks – MSI

no deadlock



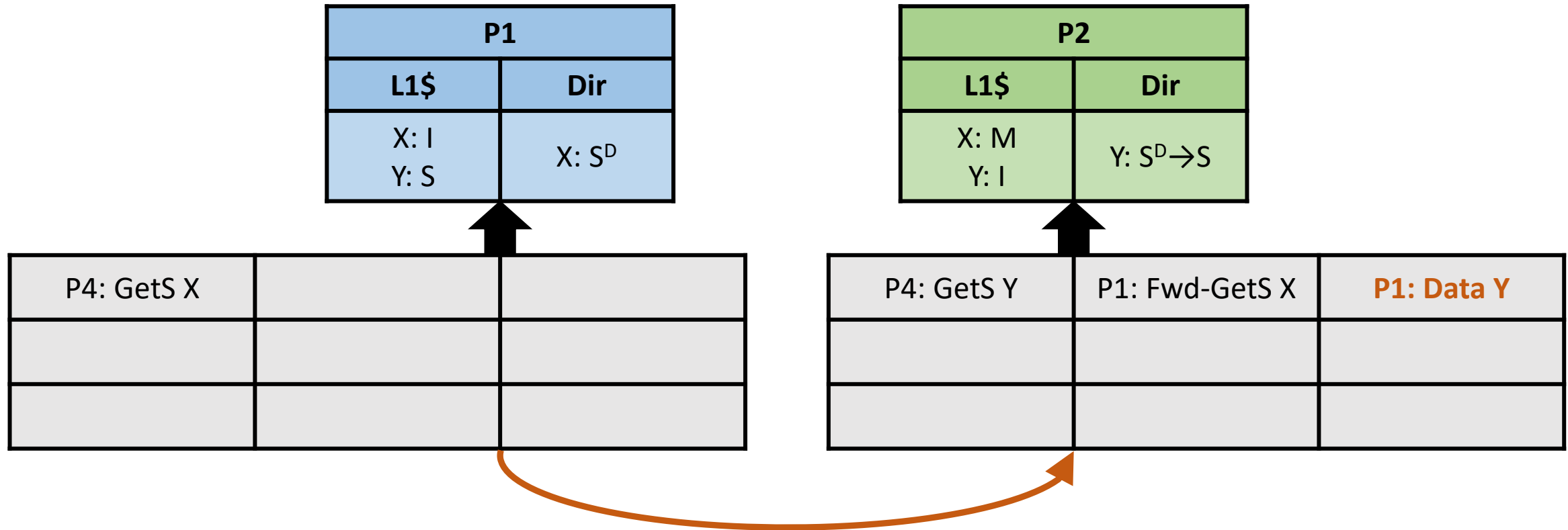
Virtual Networks – MSI

no deadlock



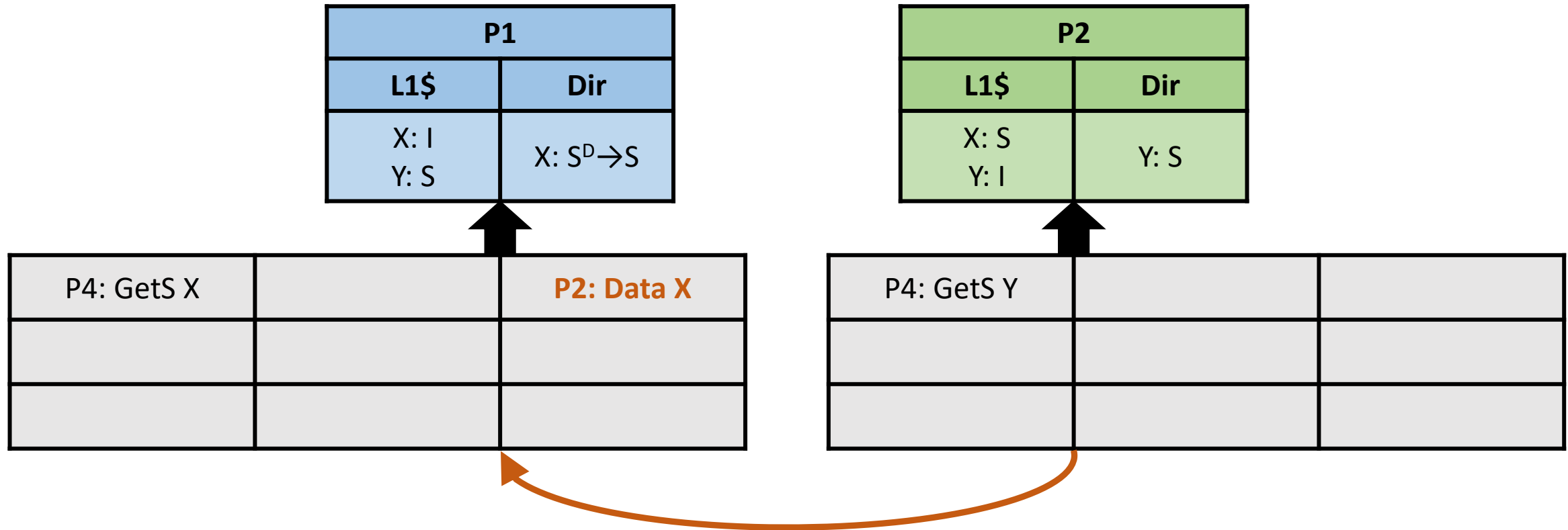
Virtual Networks – MSI

no deadlock



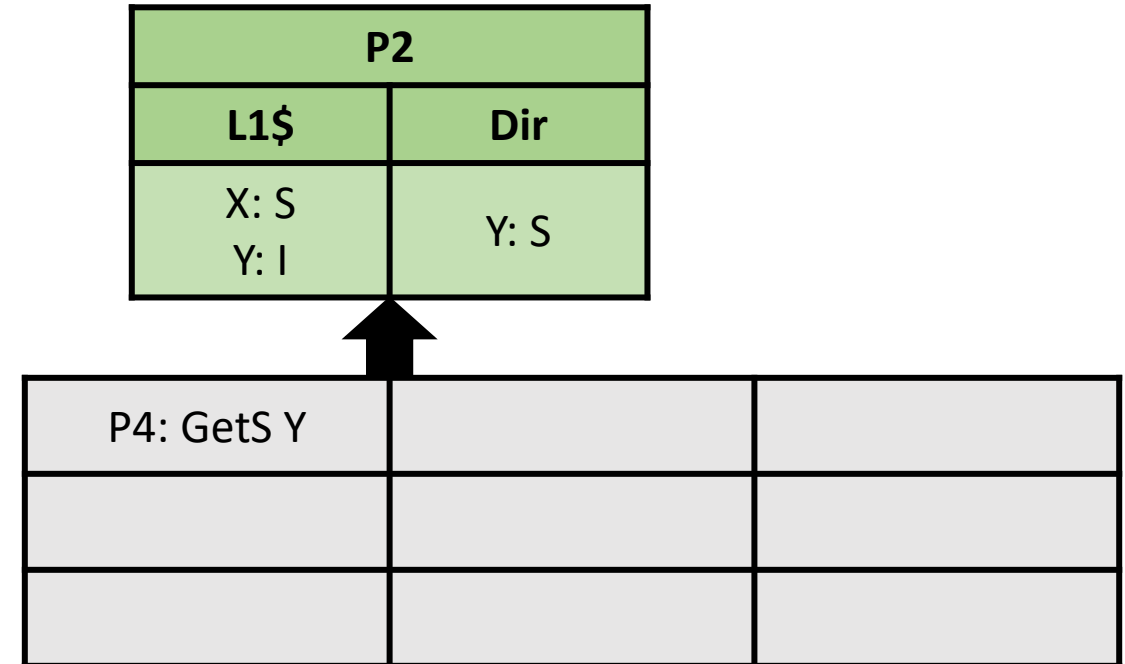
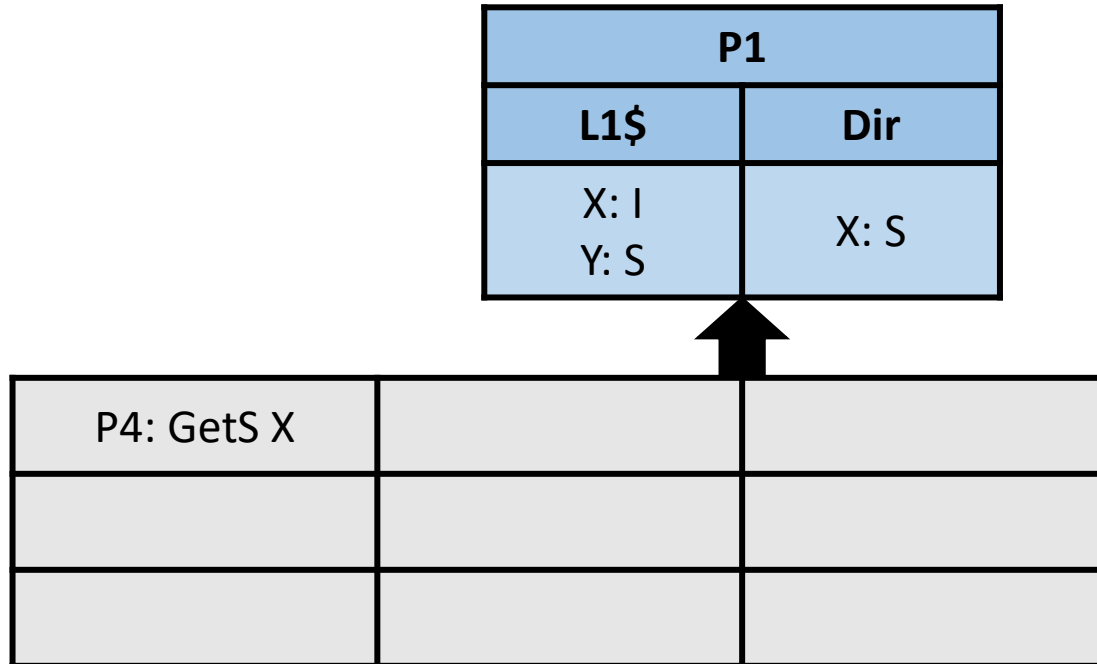
Virtual Networks – MSI

no deadlock



Virtual Networks – MSI

no deadlock



Virtual Networks – MSI

no deadlock

